

Differential Privacy

COMPCSI 752: Big Data Management

University of Auckland

(Credits to Ninh Pham)

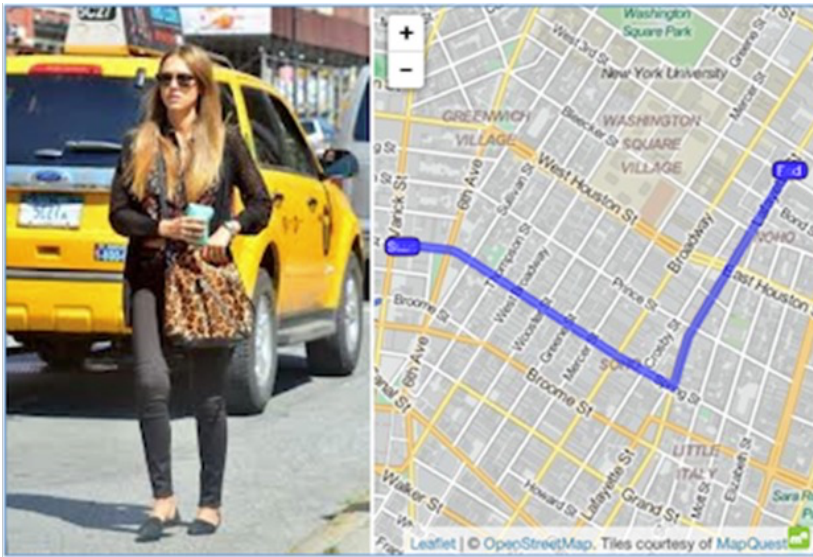
Slides are collected and edited from many resources in the reference

Auckland, May 2025

Outline

- Motivation
- Why is anonymization hard?
- Differential privacy
- Applications

Data release horror stories



Jessica Alba (actor)

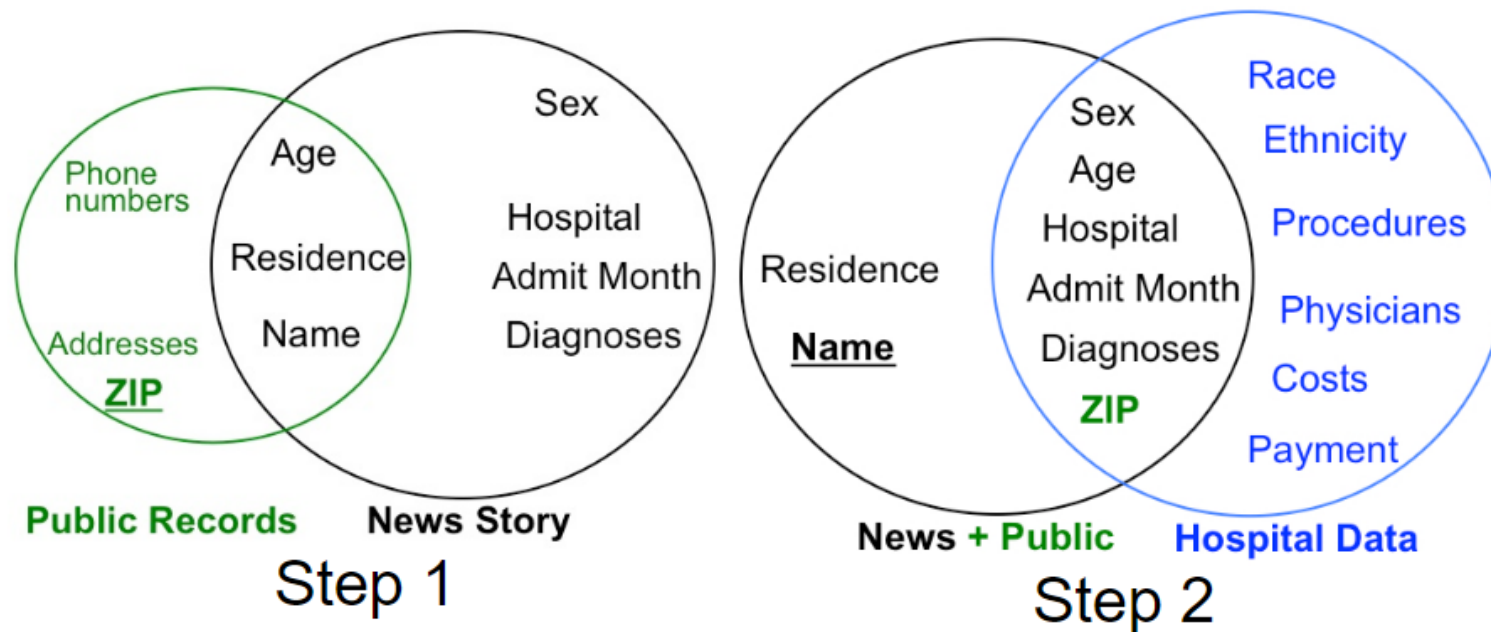
- NYC Taxi and Limousine Commission released anonymized data
- De-anonymize:
 - Google images of a star getting out of a taxi (e.g. celebrity gossip blogs)
 - Cross-reference with released data
- Alba took a taxi from Trump SoHo, and did not add a tip to \$9 fare!

<https://www.fastcompany.com/3036573/>

[nyc-taxi-data-blunder-reveals-which-celebs-dont-tip-and-who-frequents-strip-clubs](#)

Data release horror stories

Matching Known Patients to Health Records in Washington State Data



Information from news accident reports uniquely and exactly matched medical records in publicly available Washington State health data in 43% of the cases, thereby putting names to patient records.

<https://thedatamap.org/risks.html>

Data release horror stories

Strava's fitness tracker heat map reveals the location of military bases

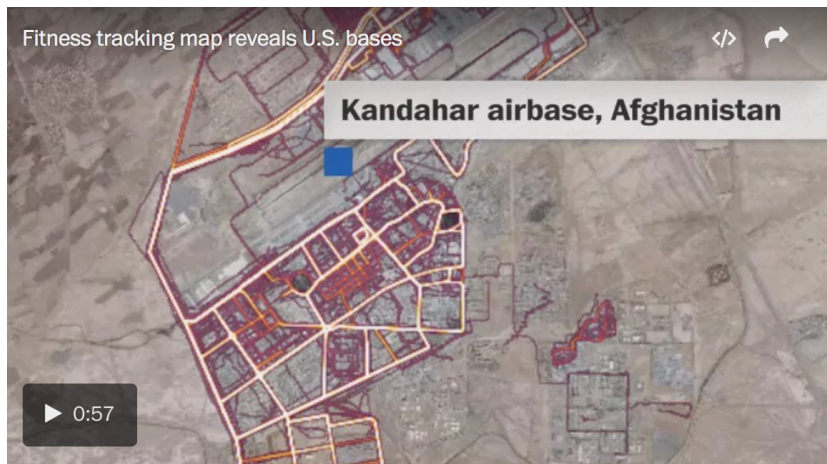
Geolocation isn't a new problem for the military

By [Andrew Liptak](#) | [@AndrewLiptak](#) | Jan 28, 2018, 3:51pm EST



**Commonwealth of Massachusetts
Group Insurance Commission**

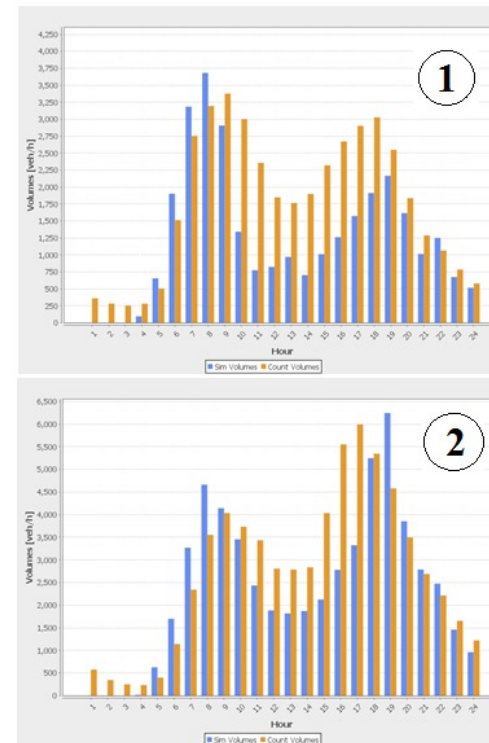
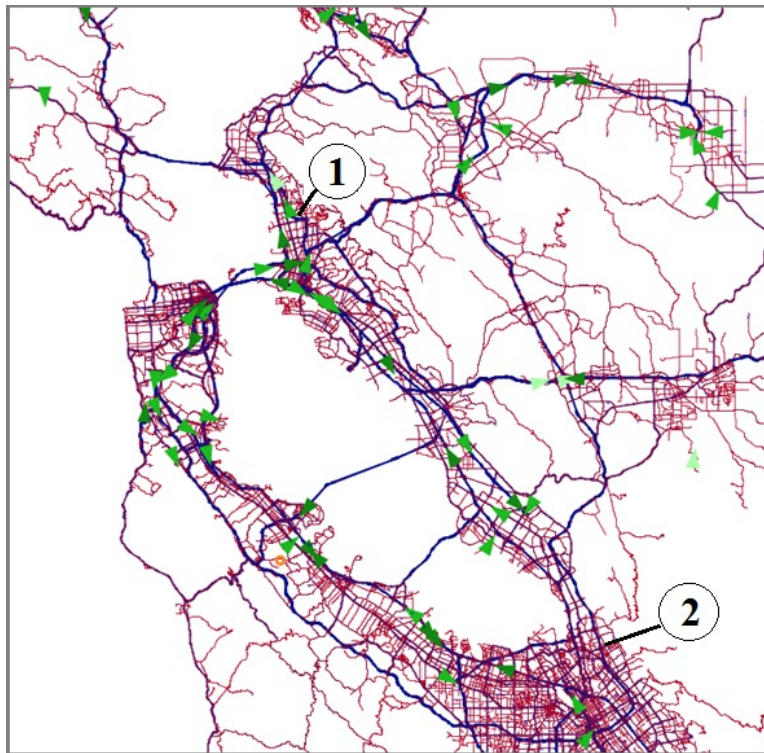
*Your
Benefits
Connection*



87% of all Americans can be uniquely identified using 3 bits of information: ZIP code, birth date, and gender.

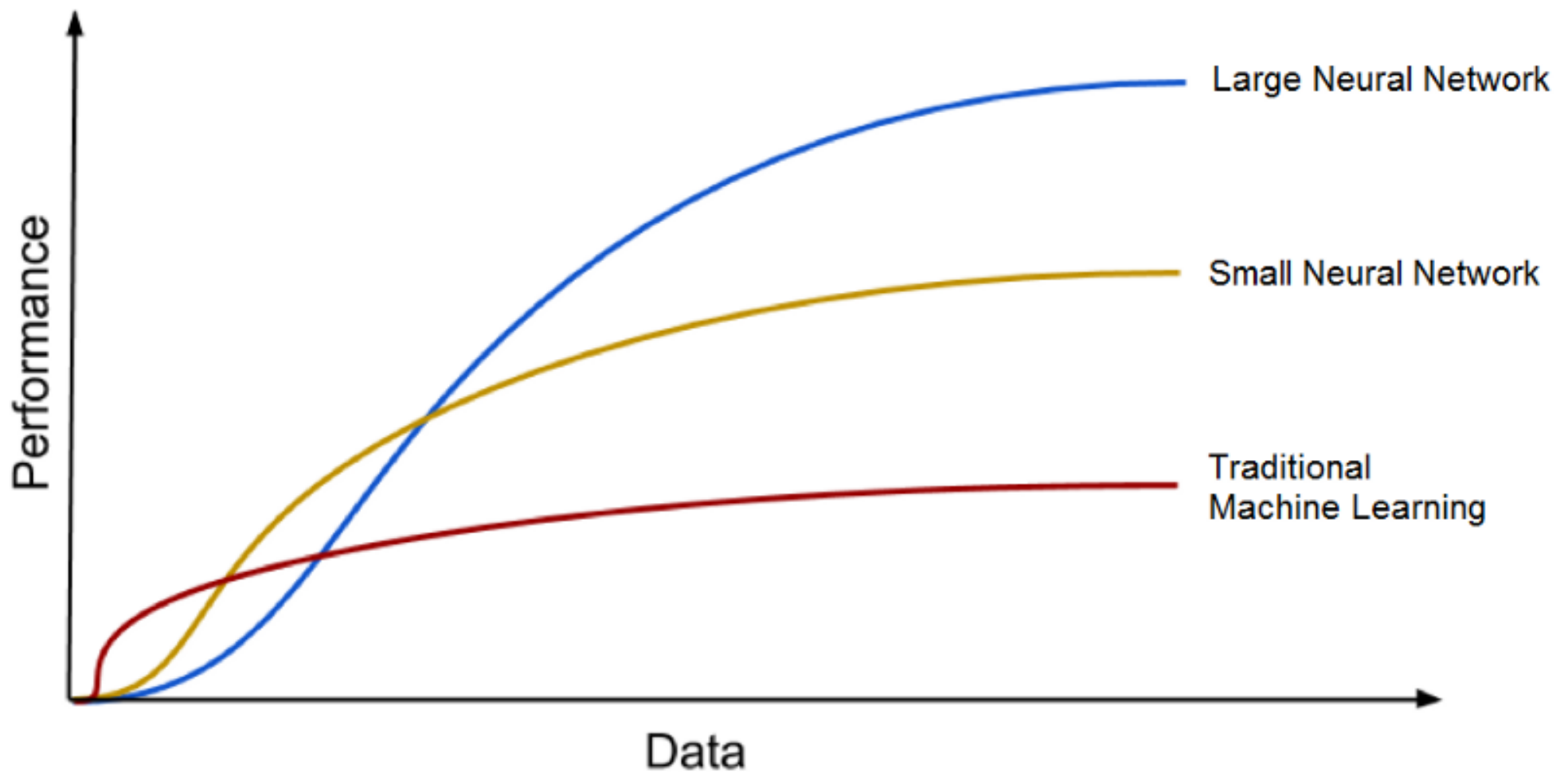
We need to solve this data release problem...

Why release data?



- Urban mobility modelling at scale
 - Smart traffic design is critical since congestion gets worse

Deep learning & big data



Big data to solve big problems

- Computation with big data:
 - Statistical correlation (genetic association)
 - Aggregate statistics (web analytics)
 - Identify events (disease outbreaks)
 - Data mining
- Individual has useful data but wants **privacy**
- How can we work on sensitive data?
 - **Easy answer:** Anonymize it and share
 - **Problem:** How to do that?



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Defence against privacy attacks

- Anonymization techniques:
 - Remove personally identifiable information
 - **K**-anonymization
- Encryption techniques:
 - Homomorphic encryption
 - Multi-party computation

Anonymization technique

- Principle:
 - Remove **identifying** information in the data before releasing
 - E.g. Sensitive columns (i.e. Name and Address) are masked
- Can we compute on anonymized data, i.e. not including personally identifiable information?

User ID	Name	Address	Account	Sub. Date
001	Alice	123 A St	Pro	01/02/20
002	Bob	234 B St	Free	02/03/21
003	Charlie	456 C St	Pro	03/04/18

User ID	Name	Address	Account	Sub. Date
001			Pro	01/02/20
002			Free	02/03/21
003			Pro	03/04/18

Linkage attack

- Netflix challenge:

- Netflix released its database for the \$1 million contest
- Sanitization: Personal identities removed

- Sparsity of data:

- No two Netflix records are similar more than 50% with high probability
- If profiles can be matched up to 50% to a profile in IMDB, then adversary knows the true identity of the profile w.h.p.

NETFLIX



Personally identifiable information (PII)

- PII definitions (from Wikipedia):
 - US: *Information which can be used to distinguish or trace an individual's identity, such as their name, social security number, biometric records, etc. alone, or when **combined** with other personal or identifying information which is linked or linkable to a specific individual, such as date and place of birth, mother's maiden name,...*
 - EU: *'personal data' shall mean any information relating to an identified or identifiable natural person ('data subject'); an identifiable person is one who can be identified, directly or indirectly, in particular by **reference** to an identification number or to one or more factors specific to his physical, physiological, mental, economic, cultural or social identity;*
- **Problem:** Without releasing entire data, how can we compute some innocent **statistics** for analysis?

Attacks from published statistics

Learning Your Identity and Disease from Research Papers: Information Leaks in Genome Wide Association Study

Rui Wang, Yong Li, XiaoFeng Wang, Haixu Tang, Xiaoyong Zhou
Indiana University Bloomington
Bloomington, IN
{wang63,yonli,xw7,hatang,zhou}@indiana.edu

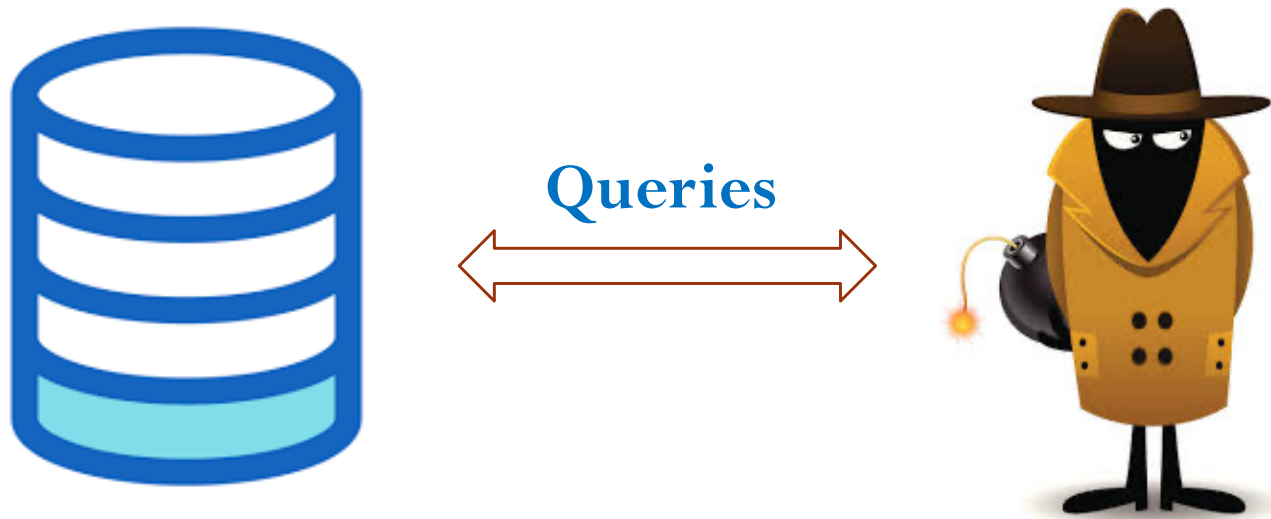
Abstract

Genome-wide association studies (GWAS) aim at discovering the association between genetic variations, particularly single-nucleotide polymorphism (SNP), and common diseases, which have been well recognized to be one of the most important and active areas in biomedical research. Also renowned is the privacy implication of such studies, which has been brought into the limelight by the recent attack proposed by Homer et al. Homer's attack demonstrates that it is possible to identify a participant of a GWAS from analyz-

1. INTRODUCTION

The rapid advancement in genome technology has revolutionized the field of human genetics by enabling the large-scale applications of genome-wide association study (GWAS) [7], a study that aims at discovering the association between human genes and common diseases. To this end, GWAS investigators determined the genotypes of two groups of participants, people with a disease (cases) and similar people without (controls) in an attempt to use statistical testing to identify genetic markers, typically single-nucleotide

Database privacy



- Need a formal privacy definition since ad-hoc solutions do not work
- How can we have privacy-preserving queries?

Blending into a crowd

- “I am safe in a group of K or more!”
 - K could vary in (3, ... 100, ... , 5.1M)
- Reasons:
 - Rare properties help re-identify someone
 - Privacy is “protection of being brought to the attention of others”



The Yale Law Journal

Volume 89, Number 3, January 1980

Privacy and the Limits of Law

Ruth Gavison[†]

Anyone who studies the law of privacy today may well feel a sense of uneasiness. On one hand, there are popular demands for increased protection of privacy, discussions of new threats to privacy, and an intensified interest in the relationship between privacy and other

Cluster-based definition

- **K-anonymity:**

- Attributes are *suppressed* or *generalized* until each row is identical to at least $K - 1$ other rows. The database is called **K**-anonymous.
- Linkage attack will result in a group of **K** records

- **Methods:**

- **Suppression:** Replace individual attributes with a *
- **Generalization:** Replace individual attributes with a broader category (e.g. office: CBD $\Rightarrow$ office: Auckland)

Why is anonymization hard?

- High-dimensional/high-resolution data is essentially unique
 - Significant computations to achieve **K**-anonymity

<i>office</i>	<i>department</i>	<i>date joined</i>	<i>salary</i>	<i>d.o.b.</i>	<i>nationality</i>	<i>gender</i>
London	IT	Apr 2015	£###	May 1985	Portuguese	Female

- Low-dimension/low-resolution is more private, but less useful in analysis

<i>office</i>	<i>department</i>	<i>date joined</i>	<i>salary</i>	<i>d.o.b.</i>	<i>nationality</i>	<i>gender</i>
UK	IT	2015	£###	1980-1985	—	Female

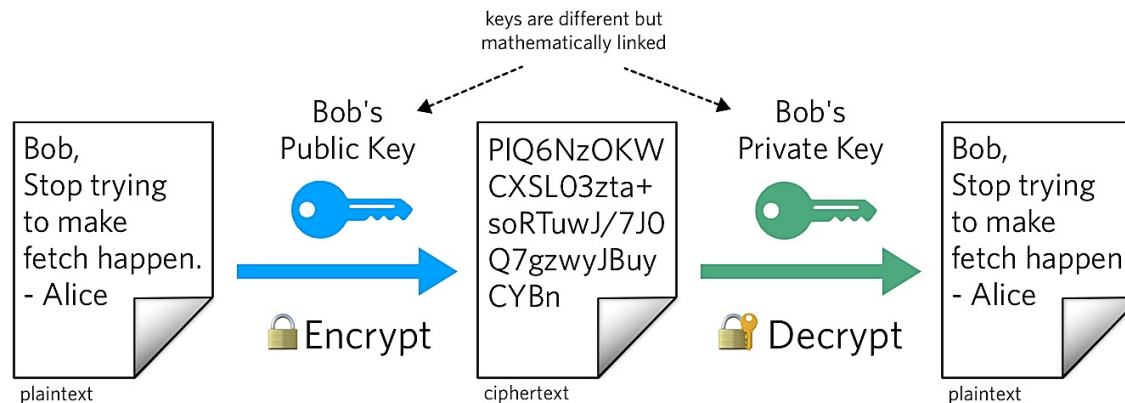
Encryption techniques

- Encryption:

- A cryptography approach converts the original representation of information (plaintext) into an alternative form (ciphertext)

- Common techniques:

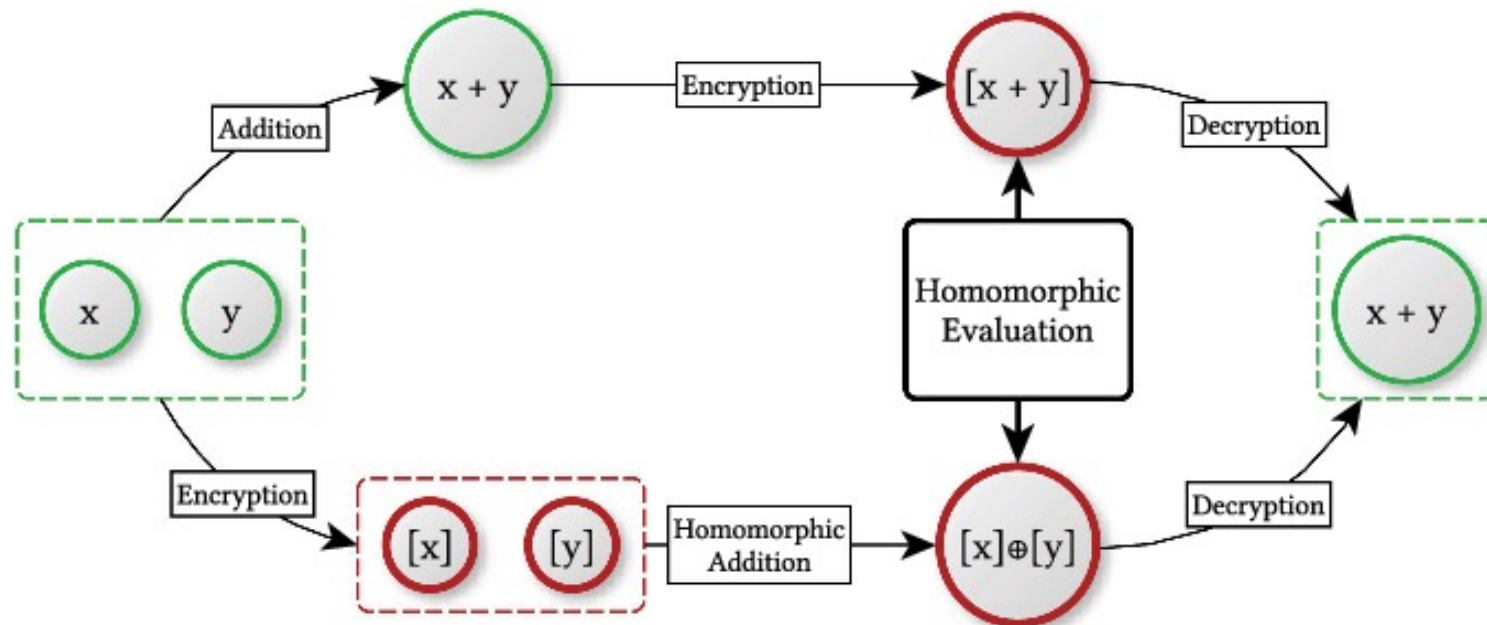
- Homomorphic encryption (HE)
- Secure multi-party computation (MPC)



Homomorphic encryption (HE)

同态

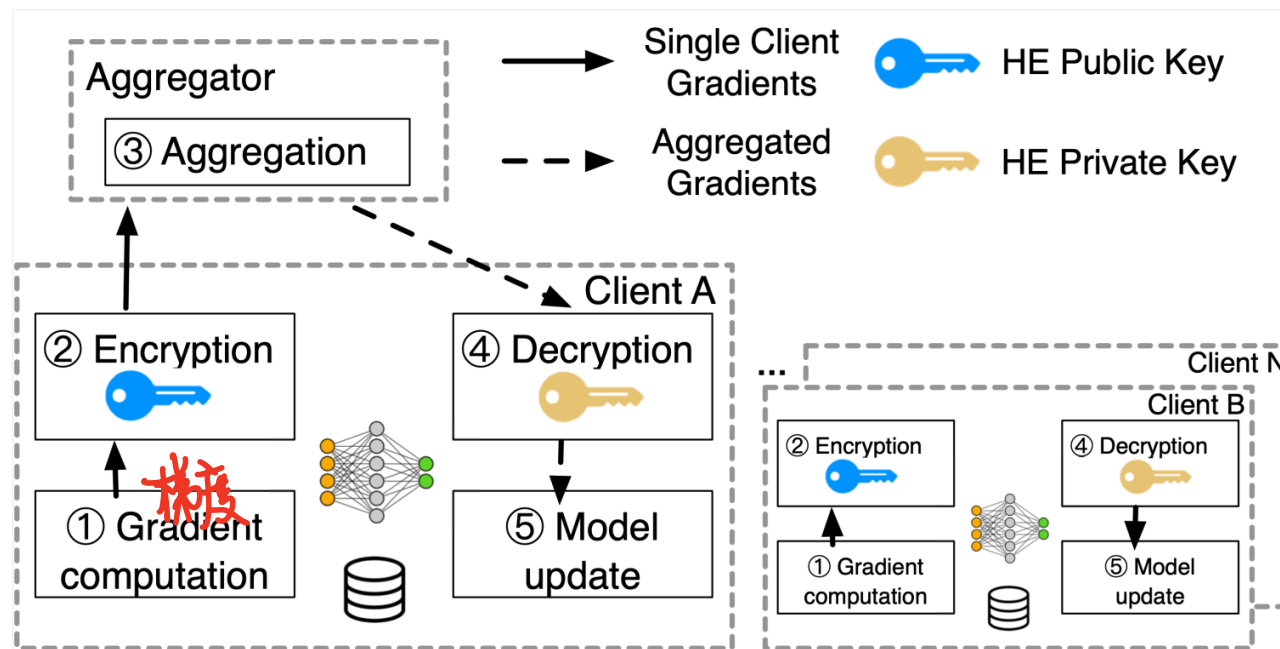
- Allow computations on encrypted data without decrypting it
 - Use a **public key** to encrypt data, and apply algebraic system (e.g. addition, multiplication) to allow computations on ciphertext
 - Only person with **matching private key** can access the decrypted results



Homomorphic encryption (HE)

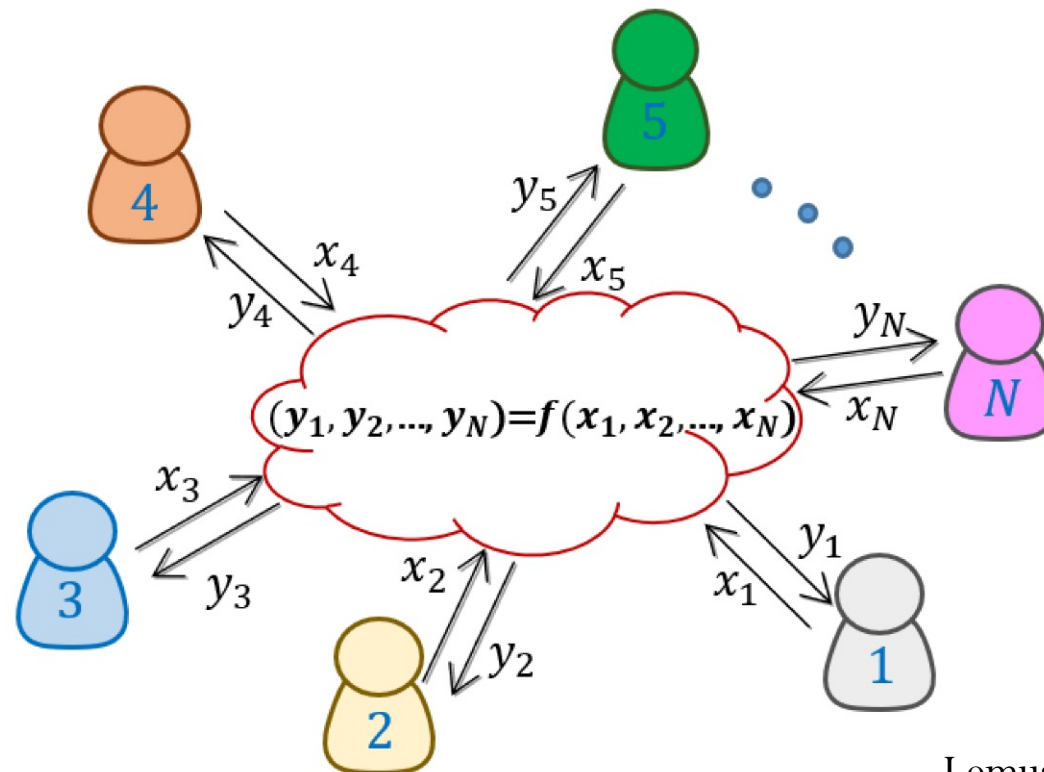
- Use cases:

- In machine learning, training data are encrypted and sent to a server (cloud) for model aggregation. Only authorized parties can access model.
- Data owner wants to use MLaaS (Machine Learning as a Service) or cloud computing services but does not trust the service provider



Secure multi-party computation (MPC)

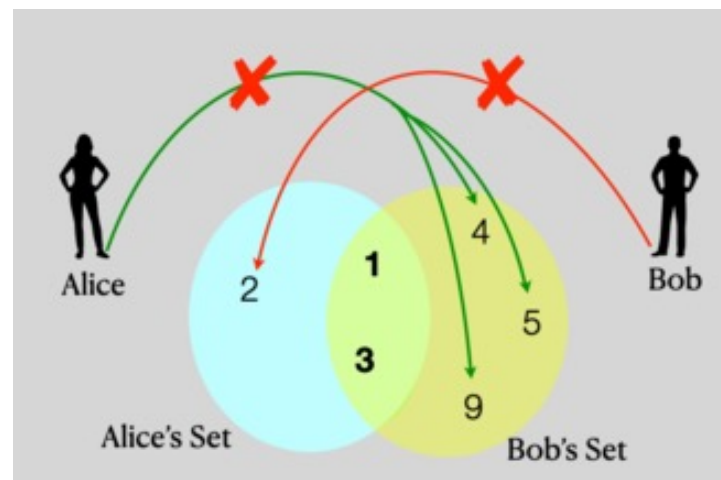
- Allow multiple parties to jointly perform computation over their private data without sharing it
 - Each party does not trust other parties or the server



Secure multi-party computation (MPC)

- Use cases:

- Query a database to get information without revealing the question asked (i.e. private information retrieval)
- Secure auction
- Insurance companies want to know the common flagged customers and their activities without sharing the whole flagged customer list
- Covid contact tracing (e.g. each user has a set of locations)



https://en.wikipedia.org/wiki/Private_set_intersection

HE vs MPC

- MPC protects the privacy of data in collaborative learning
 - Each party does not trust the others or the server
- HE keeps the data hidden from external adversaries
 - Only the data owner can decrypt the result
- Both require significant ^{expensive} computations and communication overheads and are often infeasible ^{high} for big data
 - A large number of rounds to aggregate the models
 - Significant computation on the ciphertext
 - Impractical for most of big data primitives

Privacy expectations

- Unreasonable privacy expectations:
 - *Privacy for free?* No, privatizing requires removing information (hence accuracy loss) or increasing computation
 - *Absolute privacy?* No, your neighbor's habits are correlated with your habit
- Reasonable privacy expectations:
 - *Quantitative*: Offer a knob to tune accuracy vs privacy loss
 - *Plausible deniability*: Your presence in a database cannot be ascertained
 - *Prevent targeted attacks*: Limit information leaked even in the presence of side knowledge

Privacy for free?

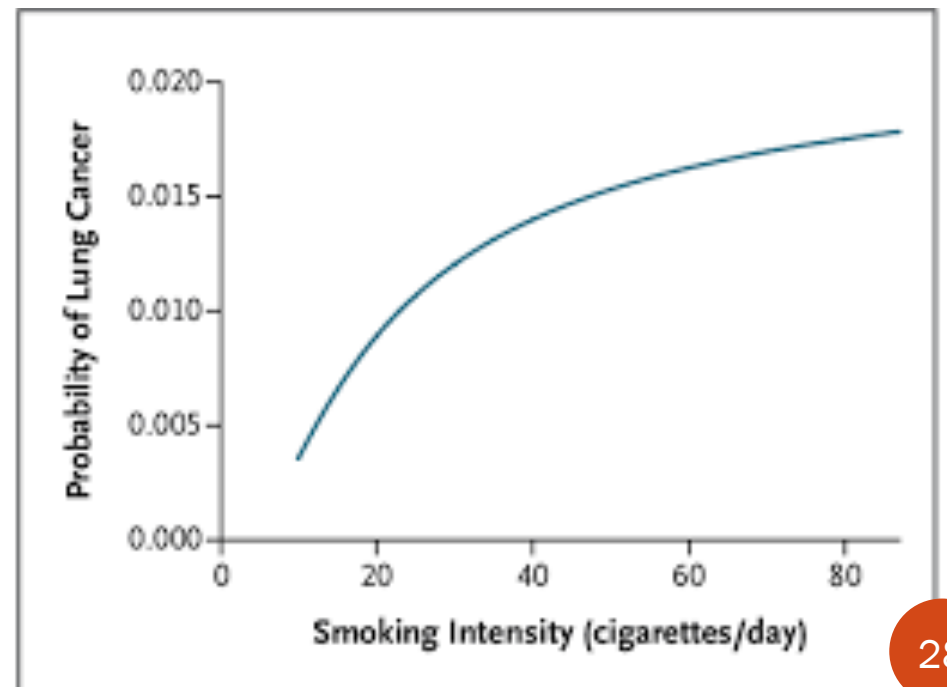
- I feel safe submitting a survey if I knew that my answer had no impact on the released results
 - $Q(D_{I-me}) = Q(D_I)$
- But... if individual answers had no impact on the released results, the results would have no utility
 - $Q(D_I) = Q(D_\emptyset)$
无意义
- Problem: Trade-off between privacy and utility
权衡 效用

Absolute privacy?

- I feel safe submitting a survey if I knew that any attacker looking at the released results **R** could not learn w.h.p. any **new** information about me personally
 - $\Pr [\text{secret}(\text{me}) \mid \mathbf{R}] = \Pr [\text{secret}(\text{me})]$
- But... if **R** shows a strong trend in my population (e.g. *everyone is age 20 – 30 and likes Taylor Swift*), w.h.p. the **trend is true of me** too (even if I do not submit a survey)
 - $\Pr [\text{secret}(\text{me}) \mid \mathbf{R}] > \Pr [\text{secret}(\text{me})]$

Absolute privacy?

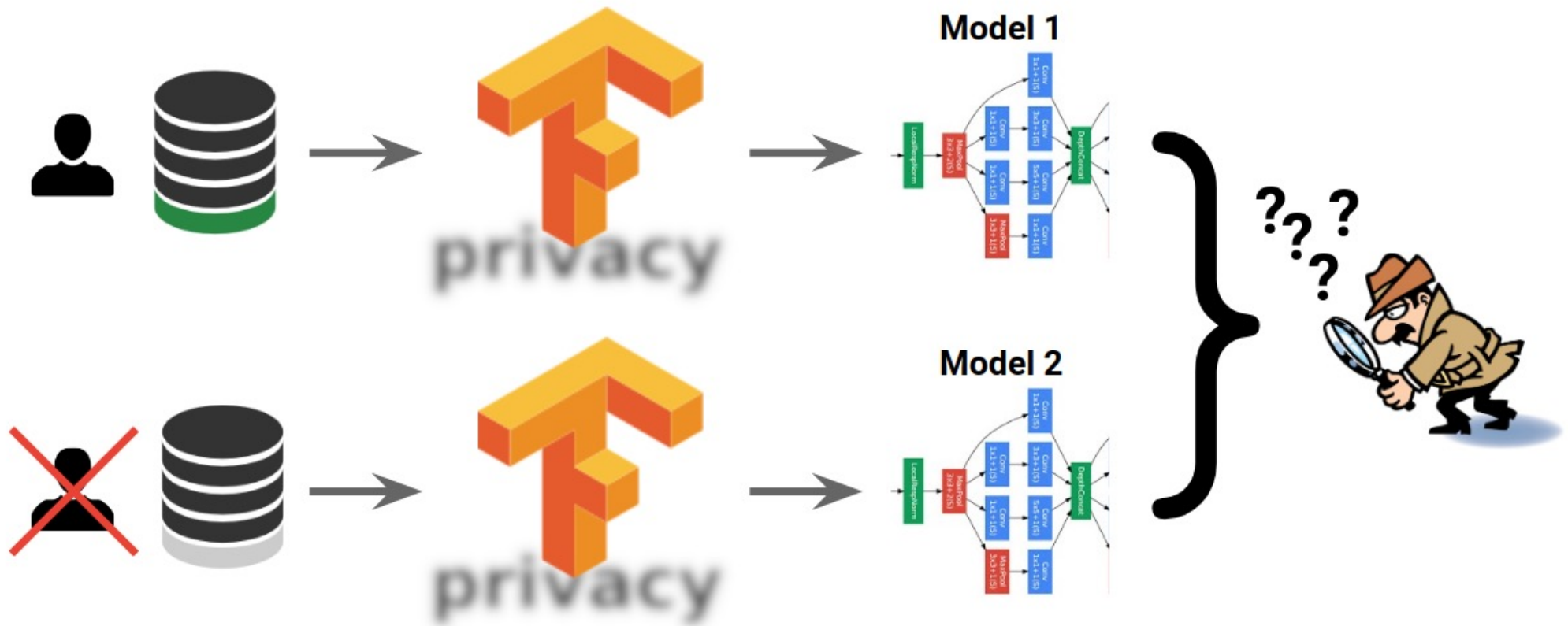
- Even worse, if an attacker knows something about me **related to** general facts about the populations, then releasing these general facts gives the attacker specific information about me (even if I do not submit a survey)
- From a survey, there is a strong correlation between smoking and being cancer
- If the insurance company knows that Alice is a smoker, it will increase her premium (even Alice had not submitted a survey)



Reasonable privacy expectations

- We cannot promise:
 - My data will not affect the results
 - An attacker could not learn new information about me, giving proper background information
- I feel safe submitting a survey...
- ... if I knew the chance that the privatized released result $\mathbf{R}$ was nearly the same, **whether or not** I submitted my information

Differential privacy



<https://blog.openmined.org/a-survey-of-differential-privacy-frameworks/>

The promise of differential privacy

- Dwork and Roth, 2014:

- *Differential privacy describes a promise, made by a data holder, or curator, to a data subject: “You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available.”*

- The 2017 Gödel Prize award to Dwork, McSherry, Nissim, Smith

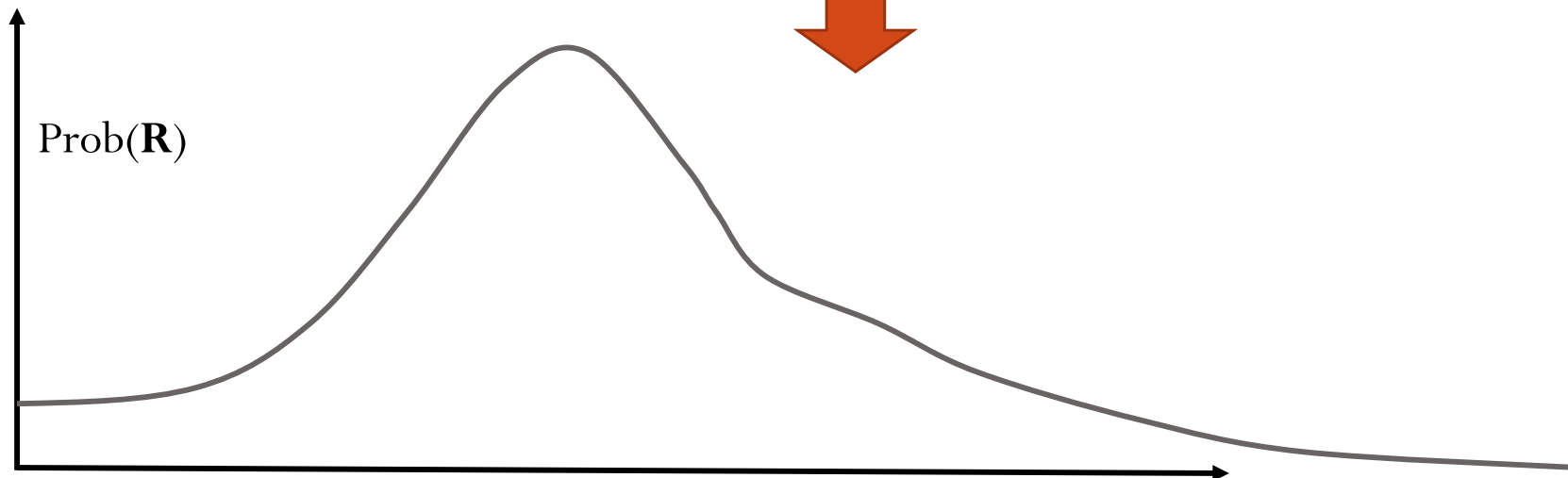
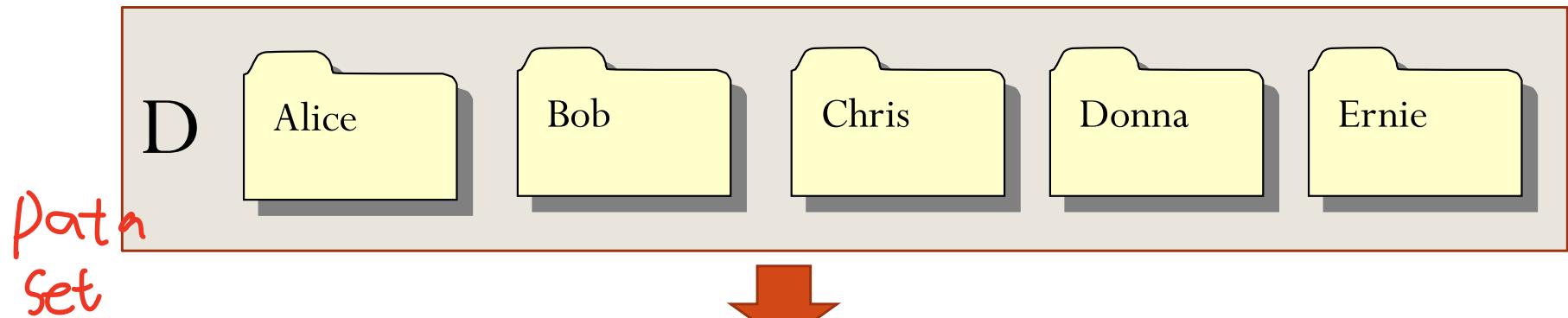
- *Differential privacy was carefully constructed to avoid numerous and subtle pitfalls that other attempts at defining privacy have faced.*
- *The intellectual impact of differential privacy has been broad, with influence on the thinking about privacy being noticeable in a huge range of disciplines, ranging from traditional areas of computer science (databases, machine learning, networking, security) to economics and game theory, false discovery control, official statistics and econometrics, information theory, genomics and, recently, law and policy.*

Outline

- Motivation
- Why is anonymization hard?
- Differential privacy (DP)
 - Definition
 - Mechanism
 - Local DP vs global DP
 - DP variants and properties
- Applications

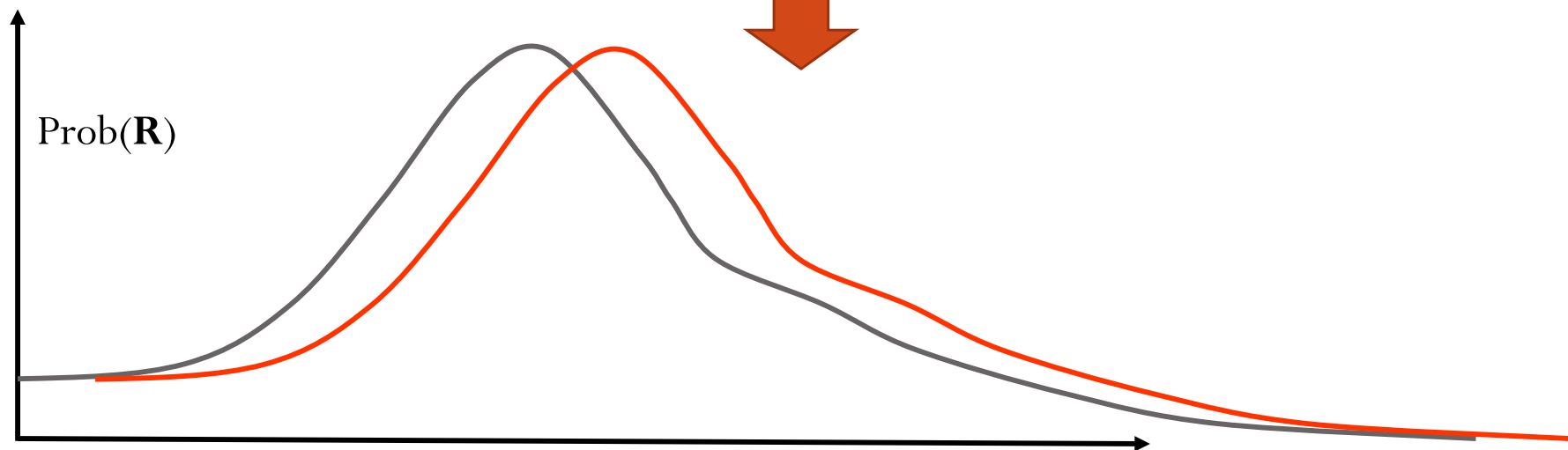
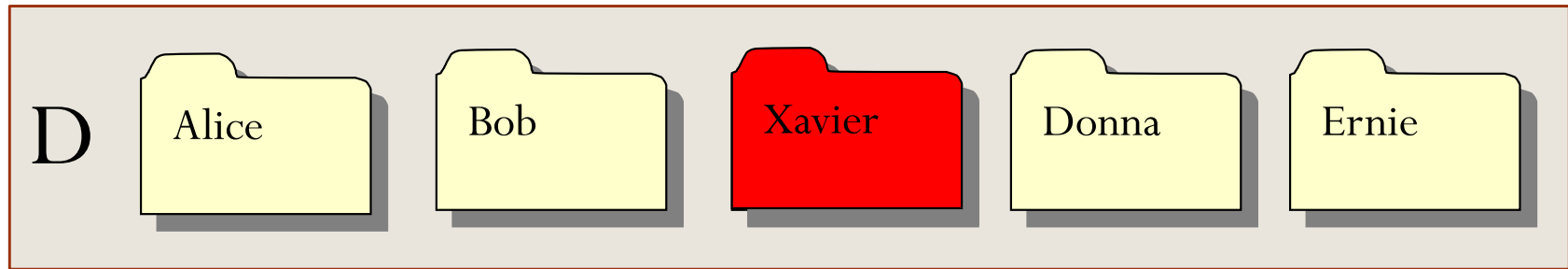
DP concept

敏感数据

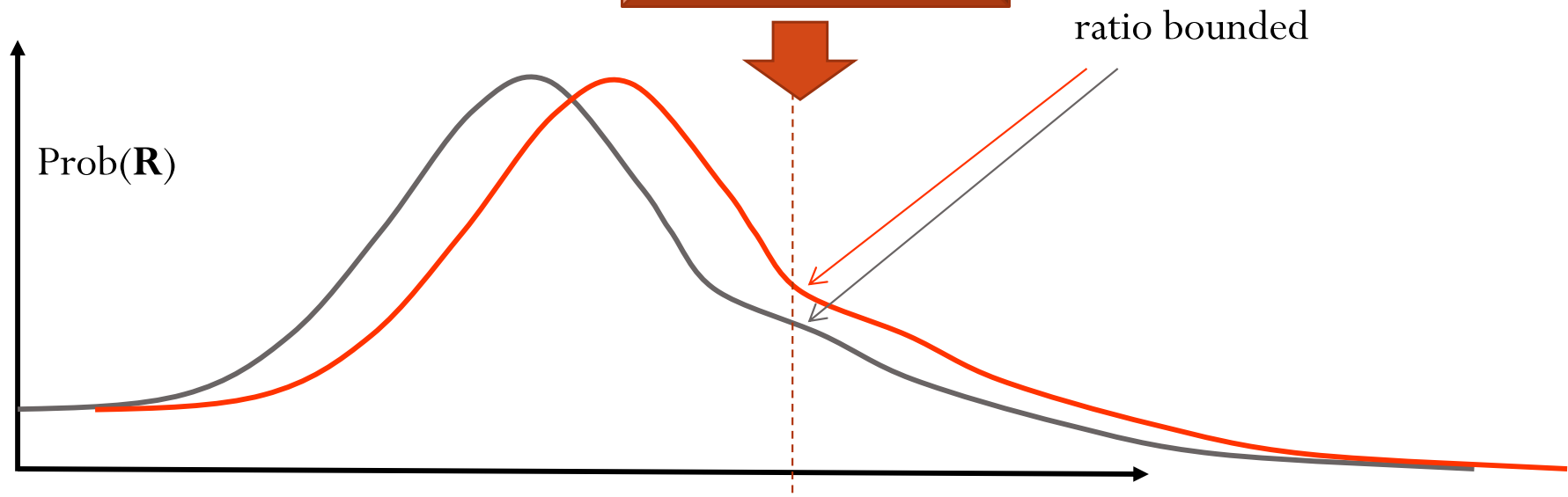
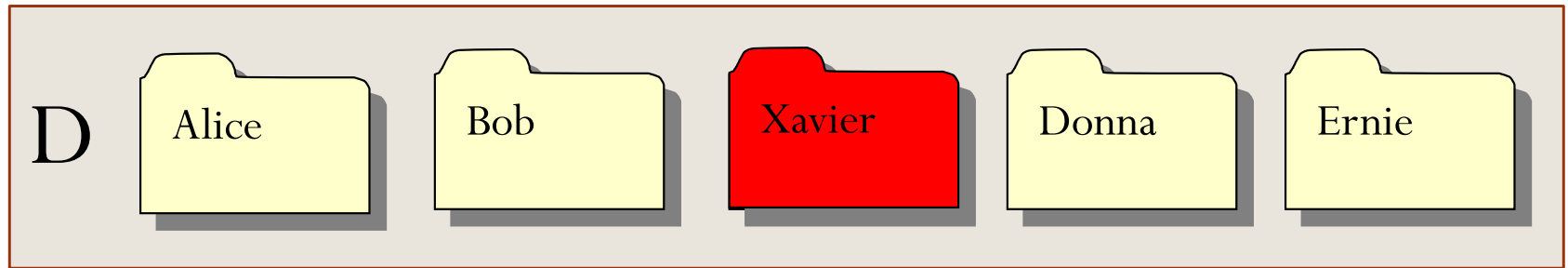


DP concept

*replace,
remove data.*



DP concept



DP definition

- $\mathbf{X}$: The data universe
- $\mathbf{D} \subset \mathbf{X}$: The data set (one element per person)

单个个体数据差异

Definition: Two datasets $\mathbf{D}, \mathbf{D}' \subset \mathbf{X}$ are *neighbors* if they differ in the data of a single individual

- The neighboring relation captures **what** is protected

DP definition

- **X**: The data universe
- **D** $\subset$ **X**: The data set (one element per person)
- Privacy parameter: $\epsilon \geq 0$

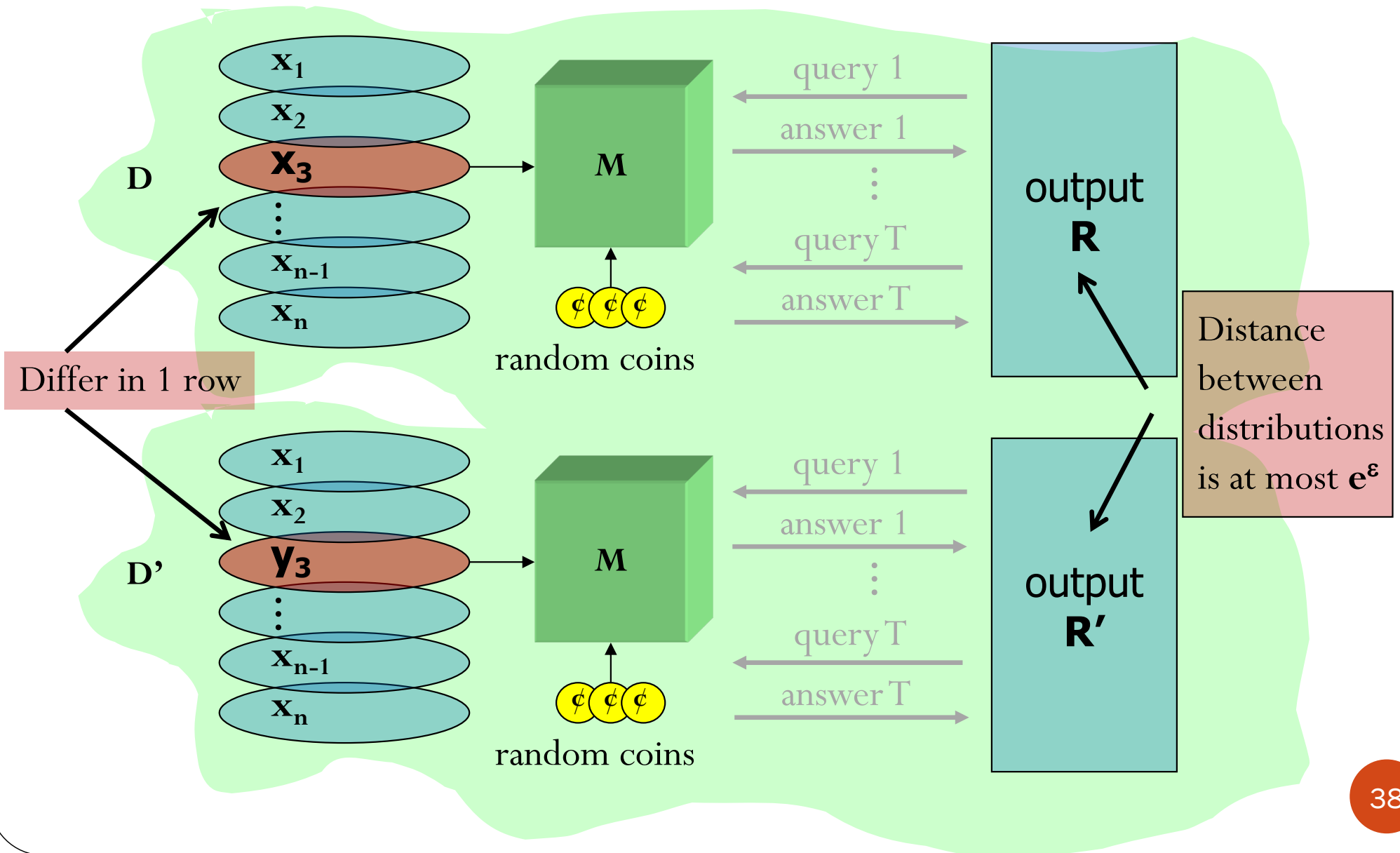
$\epsilon \uparrow$, 保护 $\downarrow$

Definition: An algorithm **M** is ϵ -differentially private if for *all* pairs of neighboring datasets **D**, **D'**, and for *all* outputs **R**:

$$\Pr [M(D) = R] \leq e^\epsilon \Pr [M(D') = R]$$

- The probability bound captures **how much** protection we get

ϵ -indistinguishable



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 - Local DP vs global DP
 - DP variants and properties
- Applications

Randomized response [Warner, 1965]

- Consider a survey of “*Have you engaged X last week?*”. The respondent is instructed to perform the following steps:
 - Flip a **first** coin
 - If **tails**, then respond truthfully
 - If **heads**, then flip a **second** coin and respond “Yes” if heads, and “No” if tails (i.e. random answer)
- Provide “plausible deniability” for the answer:
 - A response of “Yes” may be offered because the first and second coins are both **heads**, which occurs with prob. $\frac{1}{4}$
 - We achieve the privacy by the process of randomized response

Randomized response [Warner, 1965]

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 - Flip a **first** coin
 - If **tails**, then respond truthfully
 - If **heads**, then flip a **second** coin and respond “Yes” if heads, and “No” if tails (i.e. random answer)
- **Analysis:**
 - $\Pr[\text{Res} = \text{Yes} \mid \text{Truth} = \text{Yes}] = \Pr[1^{\text{st}} \text{ coin} = \text{tails}] + \Pr[1^{\text{st}} \text{ coin} = \text{heads} \ \& \ 2^{\text{nd}} \text{ coin} = \text{heads}] = \frac{1}{2} + \frac{1}{2} * \frac{1}{2} = \frac{3}{4}$
 - $\Pr[\text{Res} = \text{Yes} \mid \text{Truth} = \text{No}] = \Pr[1^{\text{st}} \text{ coin} = \text{heads} \ \& \ 2^{\text{nd}} \text{ coin} = \text{heads}] = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$
 - $\Pr[\text{Res} = \text{No} \mid \text{Truth} = \text{No}] = \frac{3}{4}$, $\Pr[\text{Res} = \text{No} \mid \text{Truth} = \text{Yes}] = \frac{1}{4}$

Randomized response [Warner, 1965]

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 - Flip a **first** coin
 - If **tails**, then respond truthfully
 - If **heads**, then flip a **second** coin and respond “Yes” if heads, and “No” if tails (i.e. random answer)
- Analysis:

$$\frac{\Pr[\text{Response} = \text{Yes} | \text{Truth} = \text{Yes}]}{\Pr[\text{Response} = \text{Yes} | \text{Truth} = \text{No}]} \quad \text{DBs differ in 1}$$

output $\nearrow$ $= \frac{3/4}{1/4} = \frac{\Pr[\text{Response} = \text{No} | \text{Truth} = \text{No}]}{\Pr[\text{Response} = \text{No} | \text{Truth} = \text{Yes}]} = 3.$

Randomized response [Warner, 1965]

- Consider a survey of “*Have you engaged X last week?*”. The respondent is instructed to perform the following steps:
 - Flip a **first** coin
 - If **tails**, then respond truthfully
 - If **heads**, then flip a **second** coin and respond “Yes” if heads, and “No” if tails (i.e. random answer)
- This mechanism is **$\ln(3)$ -DP**
 - **Specific:** We respond truthfully with prob. $\frac{3}{4}$ and falsely with prob. $\frac{1}{4}$
 - **General:** Respond truthfully with prob. $\frac{1+\epsilon}{2}$ and falsely with prob. $\frac{1-\epsilon}{2}$

Randomized response [Warner, 1965]

- Randomized response:
 - The truthful answer is $\mathbf{x} = \{0, 1\}$
 - The respondent answers truthfully ($y = \mathbf{x}$) with prob. $(1 + \epsilon)/2$ and falsely ($y = 1 - \mathbf{x}$) with prob. $(1 - \epsilon)/2$
- Randomized response is 2ϵ -DP
 - $\Pr [y = 0 | \mathbf{x} = 0] = (1 + \epsilon)/2 \approx e^\epsilon/2$
 - $\Pr [y = 0 | \mathbf{x} = 1] = (1 - \epsilon)/2 \approx e^{-\epsilon}/2$
 - $\Pr [y = 0 | \mathbf{x} = 0] / \Pr [y = 0 | \mathbf{x} = 1] \approx e^{2\epsilon}$
- Changing prob. to $e^\epsilon/(1 + e^\epsilon)$ and $1/(1 + e^\epsilon)$ yields ϵ -DP

Randomized response [Warner, 1965]

- Randomized response to estimate private mean:

- n individual answer a survey with one **binary** question
- The truthful answer for individual i is $x_i = \{0, 1\}$
- Each individual answers truthfully ($y_i = x_i$) with prob. $(1 + \epsilon)/2$ and falsely ($y_i = 1 - x_i$) with prob. $(1 - \epsilon)/2$
- Denote the mechanism by $(y_1, \dots, y_n) = \text{RR}_\epsilon(x_1, \dots, x_n)$

- Estimate the mean:

- $E[y_i] = x_i (1 + \epsilon)/2 + (1 - x_i)(1 - \epsilon)/2 = x_i \epsilon + (1 - \epsilon)/2$
- $\Rightarrow \frac{1}{n} \sum y_i = \left(\frac{1}{n} \sum x_i \right) \epsilon + (1 - \epsilon)/2$

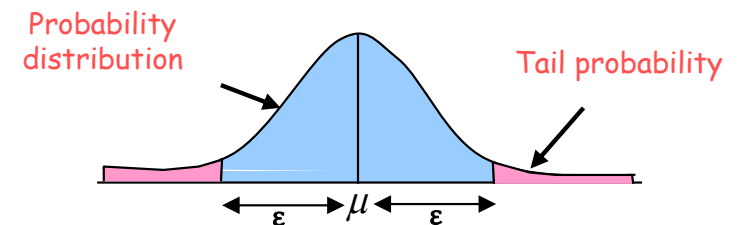
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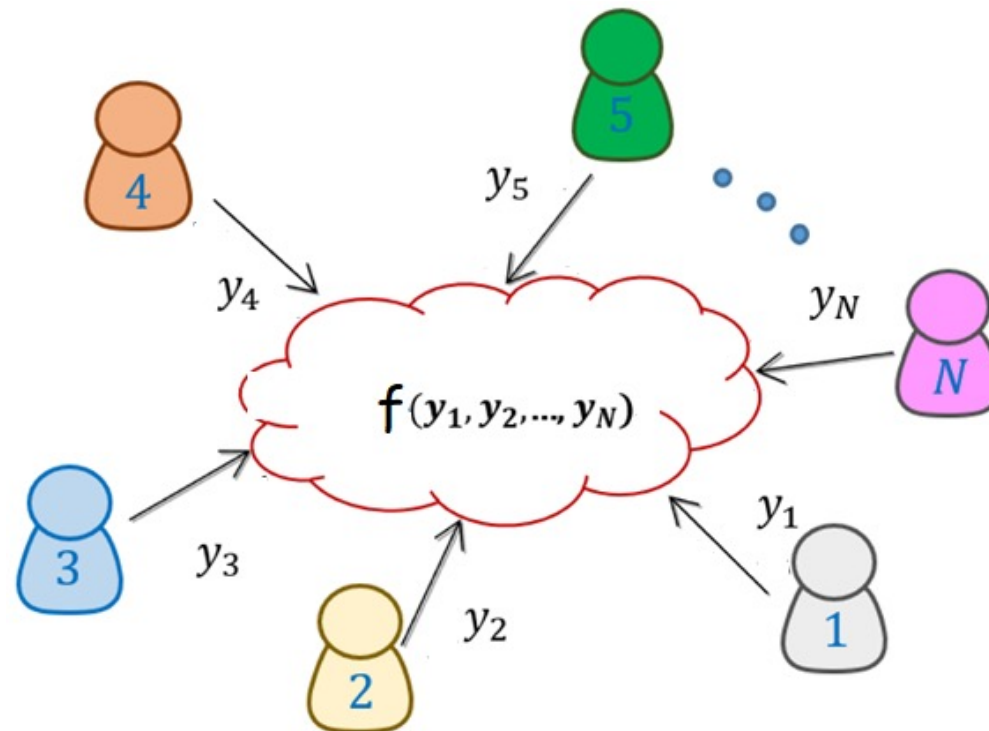
- Estimate the mean:

- $\frac{1}{n} \sum y_i = \left(\frac{1}{n} \sum x_i \right) \epsilon + (1 - \epsilon)/2$
- Chernoff bound:** $\left| \left(\frac{1}{n\epsilon} \sum y_i - \frac{1-\epsilon}{2\epsilon} \right) - \left(\frac{1}{n} \sum x_i \right) \right| \leq o\left(\frac{1}{\epsilon\sqrt{n}}\right)$ w.h.p.



Randomized response [Warner, 1965]

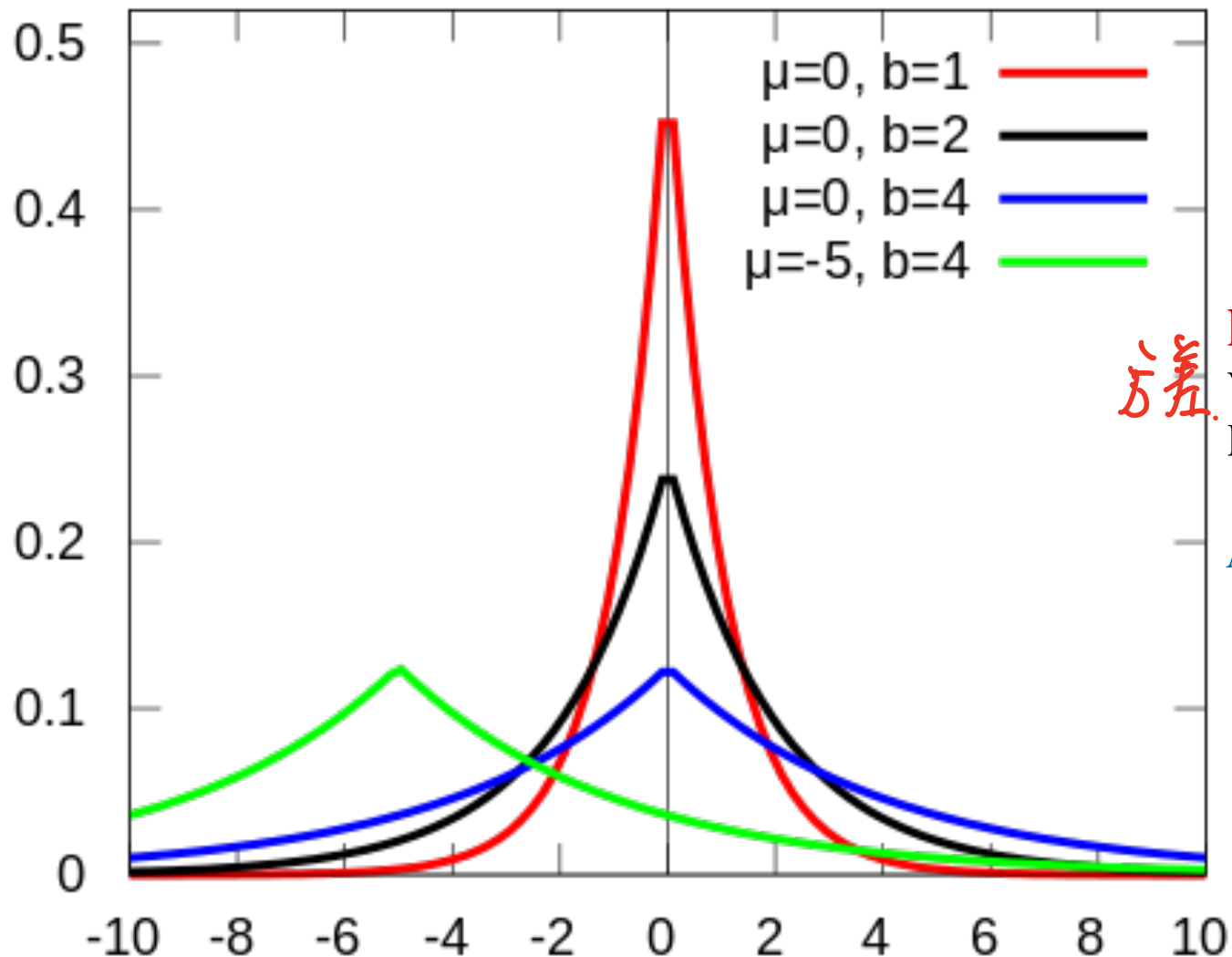
- Each individual releases their **differentially private** answer
- The curator aggregates and releases the mean estimate with low error



Laplace mechanism

- Laplace mechanism to estimate private mean:
 - A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
 - The curator proceeds by
 - Compute the mean $\mu = (\mathbf{x}_1 + \dots + \mathbf{x}_n) / \mathbf{n}$
 - Sampling noise $\mathbf{Z} \sim \text{Lap}(0, 1/\epsilon \mathbf{n})$
 - Release the noisy mean $\tilde{\mu} = \mu + \mathbf{Z}$
 - Denote the mechanism by $\tilde{\mu} = M_{\text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$
- M_{Lap} is ϵ -DP

Laplace distribution $\text{Lap}(\mu, b)$



$b \uparrow$ 保护越强
准确性少

$p(z) = \exp(-|z - \mu|/b)/2b$
方差
variance = $2b^2$
Increasing b flattens curve

Abuse of notation:

$\text{Lap}(b) = \text{Lap}(0, b)$

↓
默认 $\mu=0$,

Laplace mechanism

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 - A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
 - The curator proceeds by
 - Compute the mean $\boldsymbol{\mu} = (\mathbf{x}_1 + \dots + \mathbf{x}_n) / \mathbf{n}$
 - Sampling noise $\mathbf{Z} \sim \text{Lap}(1/\epsilon \mathbf{n})$
 - Release the noisy mean $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} + \mathbf{Z}$
 - Denote the mechanism by $\tilde{\boldsymbol{\mu}} = \mathbf{M}_{\text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$
- $\mathbf{M}_{\text{Lap}}$ is ϵ -DP
 - Given two different databases $\mathbf{D}, \mathbf{D}'$ different from 1, then
$$|\boldsymbol{\mu}_{\mathbf{D}} - \boldsymbol{\mu}_{\mathbf{D}'}| = 1/\mathbf{n}$$
 - Compute $\Pr[\mathbf{M}_{\text{Lap}}(\mathbf{D}) = \mathbf{a}] / \Pr[\mathbf{M}_{\text{Lap}}(\mathbf{D}') = \mathbf{a}]$ for any outcome $\mathbf{a}$

Laplace mechanism

- Laplace mechanism to estimate private mean:

- A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
- The curator proceeds by

- Compute the mean $\mu = (\mathbf{x}_1 + \dots + \mathbf{x}_n) / n$
- Sampling noise $\mathbf{Z} \sim \text{Lap}(1/\epsilon n)$
- Release the noisy mean $\tilde{\mu} = \mu + \mathbf{Z}$

$$\exp(1) = e$$

$\mathbf{Z}$ and $\tilde{\mu}$ are continuous variables

- Denote the mechanism by $\tilde{\mu} = M_{\text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$

- M_{Lap} is ϵ -DP

$$\text{Lap}(b) : p(z) = \exp(-|z|/b)/2b$$

$$\begin{aligned} \frac{\Pr[\mu_D + Z = a]}{\Pr[\mu_{D'} + Z' = a]} &= \frac{\Pr[Z = a - \mu_D]}{\Pr[Z' = a - \mu_{D'}]} = \frac{\exp(-\epsilon n |a - \mu_D|) * \epsilon n / 2}{\exp(-\epsilon n |a - \mu_{D'}|) * \epsilon n / 2} \\ &= \exp(\epsilon n (|a - \mu_{D'}| - |a - \mu_D|)) = \exp\left(\epsilon n * \frac{1}{n}\right) = e^\epsilon \end{aligned}$$

$$|\mu_D - \mu_{D'}| = 1/n$$

Laplace mechanism

- Laplace mechanism to estimate private mean:
 - A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
 - The curator proceeds by
 - Compute the mean $\boldsymbol{\mu} = (\mathbf{x}_1 + \dots + \mathbf{x}_n) / \mathbf{n}$
 - Sampling noise $\mathbf{Z} \sim \text{Lap}(1/\epsilon \mathbf{n})$
 - Release the **noisy** mean $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} + \mathbf{Z}$
 - Denote the mechanism by $\tilde{\boldsymbol{\mu}} = \mathbf{M}_{\text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$
- $\mathbf{M}_{\text{Lap}}$ is ϵ -DP
 - $\mathbf{E} [\tilde{\boldsymbol{\mu}}] = \boldsymbol{\mu}$ since $\mathbf{E}[\mathbf{Z}] = 0$
 - Tail bound of $\text{Lap}(1/\epsilon \mathbf{n})$: $|\tilde{\boldsymbol{\mu}} - \boldsymbol{\mu}| \leq \mathbf{O}\left(\frac{1}{\epsilon \mathbf{n}}\right)$ w.h.p.

 If $\mathbf{Z} \sim \text{Lap}(b)$ and $t > 0$, $\Pr [|\mathbf{Z}| \geq t.b] \leq \exp(-t)$

Other private computing

- Examples of private mean estimate in $\{0, 1\}$:
 - What is the percentage of passed students in CS 752-2023?
 - What is the average female survey taker?
- Variants of private mean estimate in real-value:
 - What is the average of grades of CS 752-2023?
 - What is the average income of Auckland?
- Private counting/sum in real-value:
 - What is the number of passed student in CS 752-2023?
 - In total, how many Taylor Swift albums are bought by survey takers?

Laplace mechanism

- Laplace mechanism to estimate private mean:
 - A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
 - The curator proceeds by
 - Compute the mean $\mu = (\mathbf{x}_1 + \dots + \mathbf{x}_n) / n$
 - Sampling noise $\mathbf{Z} \sim \text{Lap}(1/\epsilon n)$
 - Release the noisy mean $\tilde{\mu} = \mu + \mathbf{Z}$
 - Denote the mechanism by $\tilde{\mu} = M_{\text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$

- M_{Lap} is ϵ -DP

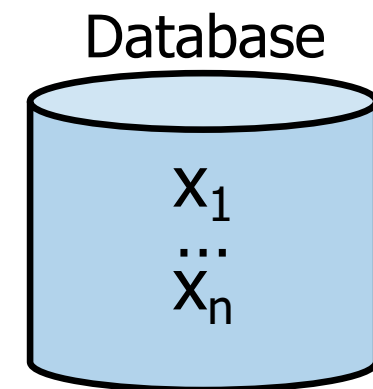
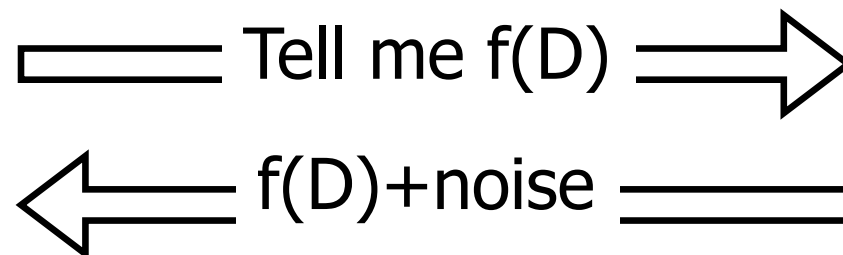
$$\frac{\Pr[\mu_D + Z = a]}{\Pr[\mu_{D'} + Z' = a]} = \frac{\Pr[Z = a - \mu_D]}{\Pr[Z' = a - \mu_{D'}]} = \frac{\exp(-\epsilon n |a - \mu_D|) * \epsilon n / 2}{\exp(-\epsilon n |a - \mu_{D'}|) * \epsilon n / 2}$$

$$= \exp(\epsilon n (|a - \mu_{D'}| - |a - \mu_D|)) = \exp\left(\epsilon n * \frac{1}{n}\right) = e^\epsilon$$

Lap(b) : $p(z) = \exp(-|z|/b)/2b$

Since $|\mu_D - \mu_{D'}| = 1/n$, Lap($1/\epsilon n$) suffices

Laplace mechanism



- **Intuition:** $f(\mathbf{x})$ can be released accurately when f is **insensitive** to individual entries $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}$
- **Global sensitivity** $GS_f = \max_{\text{neighbors } D, D'} \|f(D) - f(D')\|_1$
 - Example: $GS_{\text{average}} = 1/n$ for sets of bits
- **Theorem:** $f(\mathbf{x}) + \text{Lap}(GS_f / \epsilon)$ is ϵ -DP
 - Noise generated from Laplace distribution

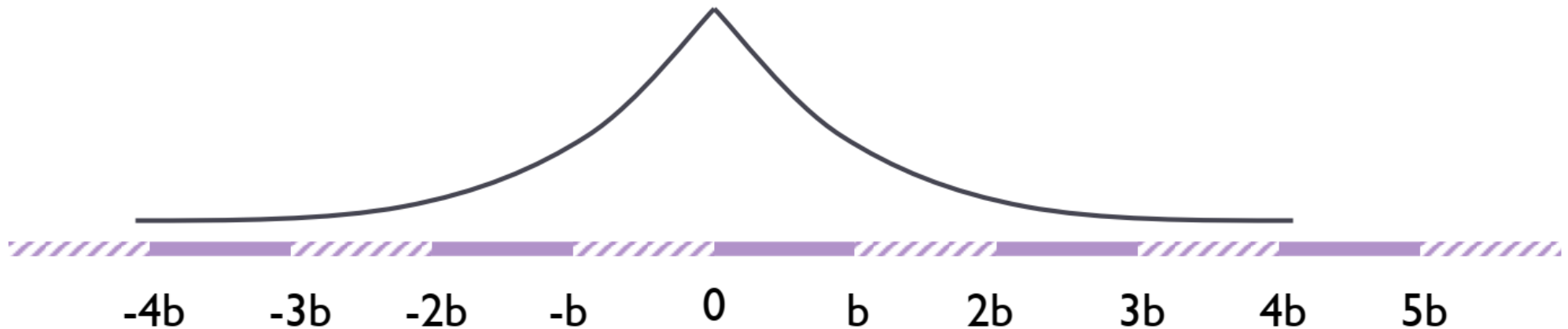
Lipschitz
constant of f

Laplace mechanism

- Private mean/sum estimate in real-value:
 - What is the average of grades of CS 752-2023?
 - **GS = 10/200** (if scale = 10 and #students = 200)
 - What is the number of passed student in CS 752-2023?
 - **GS = 1**
 - What is the total yearly income of Auckland?
 - **GS = 560,000** (i.e. the max personal income in Auckland)
 - In total, how many Taylor Swift albums are bought by survey takers?
 - **GS = 11** (Taylor has 11 released albums)

Laplace mechanism

- Global sensitivity: $\mathbf{GS}_f = \Delta f = \max_{\text{neighbors } D, D'} \|f(D) - f(D')\|_1$



Noise depends on f and ϵ , not on the database

Smaller sensitivity (Δf) means less distortion

Output perturbation mechanism

- Global sensitivity:

- For any function $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{R}^d$, define $\Delta \mathbf{f} = \max_{\text{neighbors } D, D'} \|\mathbf{f}(D) - \mathbf{f}(D')\|_1$

- The Laplace mechanism is an example of a general class of mechanisms where $\mathbf{f} = (\mathbf{x}_1 + \dots + \mathbf{x}_n)/n$ and $\mathbf{x}_i$ is a real

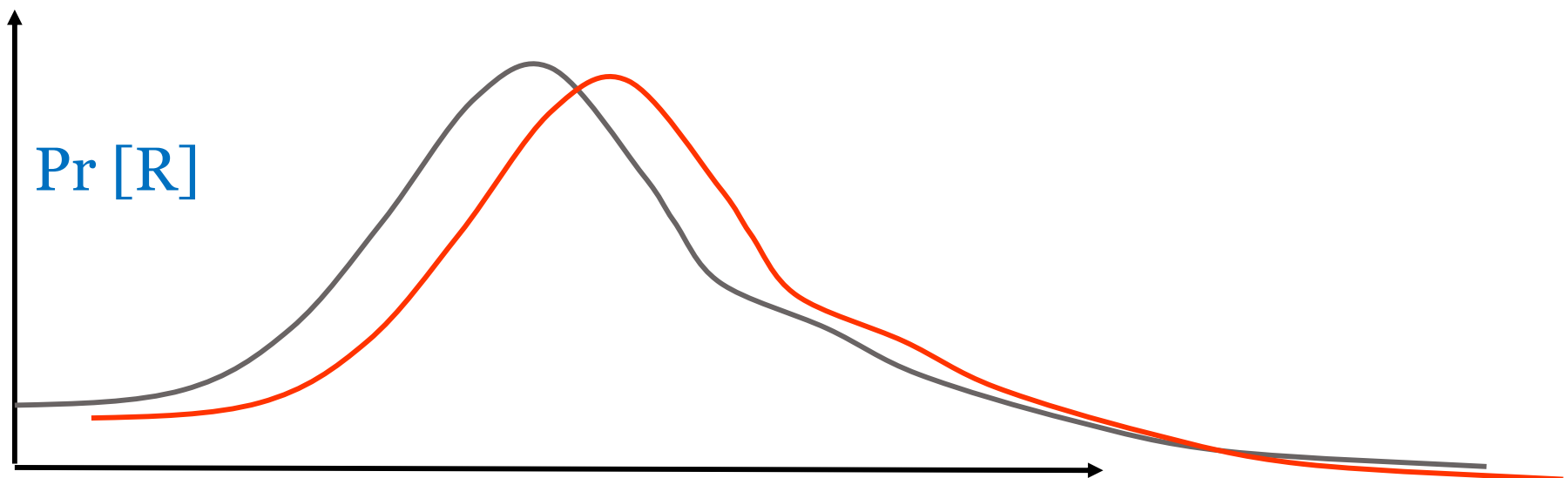
Output perturbation mechanism

- Global sensitivity:
 - For any function $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{R}^d$, define $\Delta \mathbf{f} = \max_{\text{neighbors } \mathbf{D}, \mathbf{D}'} \|\mathbf{f}(\mathbf{D}) - \mathbf{f}(\mathbf{D}')\|_1$
- Output perturbation $\mathbf{M}_{\mathbf{f}, \text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$:
 - A **trusted** curator holds one vector $\mathbf{x}_i \in \mathbf{R}^d$ for each of n individuals
 - The curator proceeds by
 - Compute the function $\mathbf{f}(\mathbf{x}_1, \dots, \mathbf{x}_n)$ of the data
 - Sampling a noise vector $\mathbf{Z} \sim \text{Lap}(\Delta \mathbf{f} / \epsilon)^d$
 - Release the **noisy** vector $\mathbf{f}(\mathbf{x}_1, \dots, \mathbf{x}_n) + \mathbf{Z}$
- $\mathbf{M}_{\mathbf{f}, \text{Lap}}(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is ϵ -DP

Differential Privacy: Summary

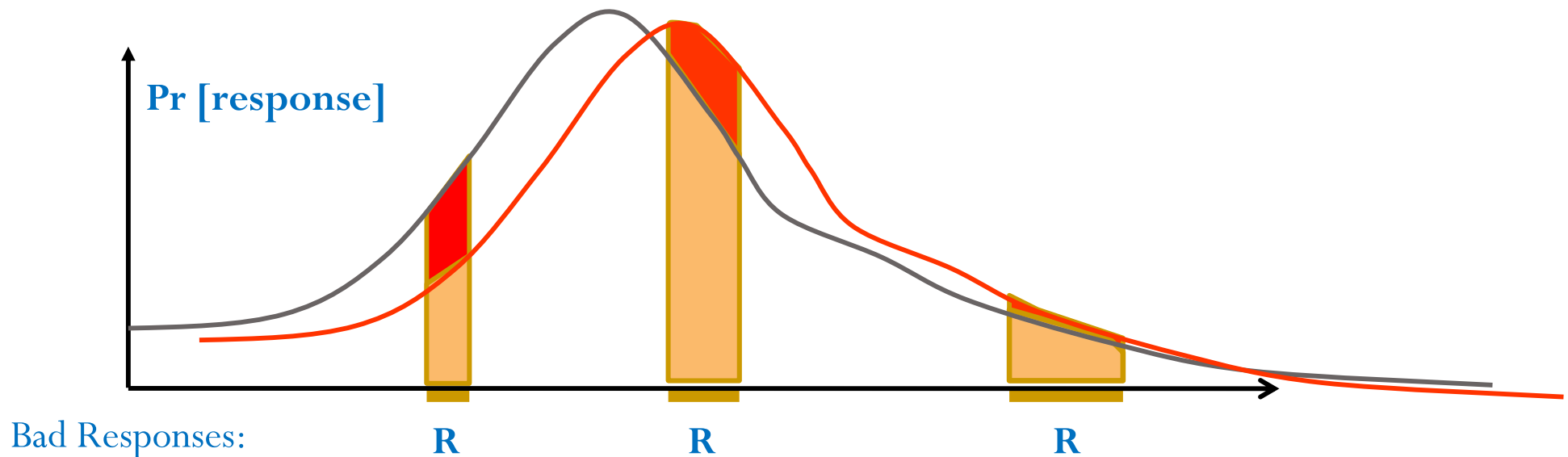
- $\mathcal{M}$ gives ϵ -differential privacy if for all values of DB and Me and all responses R :

$$1 - \epsilon \approx e^{-\epsilon} \leq \frac{\Pr[\mathcal{M}(DB) = R]}{\Pr[\mathcal{M}(DB + Me) = R]} \leq e^{\epsilon} \approx 1 + \epsilon$$



Promise of DP

- No perceptible risk is incurred by joining DB
- Anything adversary can do to me, it could do without me



Outline

- Motivation
- Why is anonymization hard
- Differential privacy (DP)
 - Definition
 - Mechanism
 - Local DP vs global DP
 - DP variants and properties
- Applications

Randomized response vs Laplace

- Randomized response to estimate private mean:

- Given the truthful answer for individual i is $x_i = \{0, 1\}$, and $\mu = \frac{1}{n} \sum x_i$
- Individual i answers truthfully ($y_i = x_i$) with prob. $(1 + \epsilon)/2$ and falsely ($y_i = 1 - x_i$) with prob. $(1 - \epsilon)/2$
- Release an unbiased estimate $\tilde{\mu}_R = \frac{1}{n\epsilon} \sum y_i - \frac{1-\epsilon}{2\epsilon}$
- $|\tilde{\mu}_R - \mu| \leq \mathcal{O}\left(\frac{1}{\epsilon\sqrt{n}}\right)$

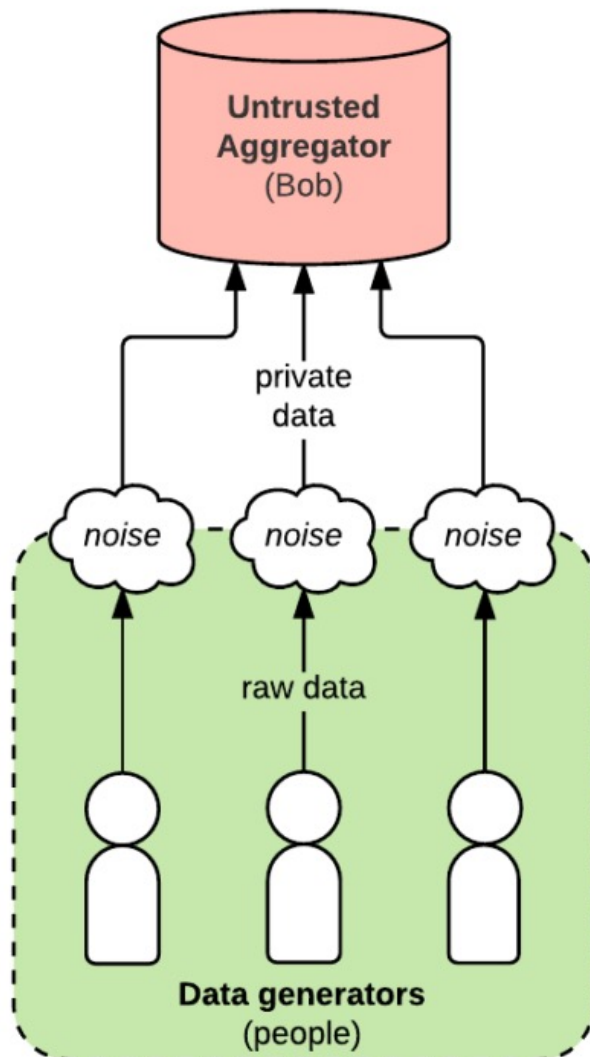
- Laplace mechanism to estimate private mean:

- A **trusted** curator holds one bit $x_i = \{0, 1\}$ for each of n individuals
- The curator compute the mean μ
- Release an unbiased estimate $\tilde{\mu}_L = \mu + Z$ where $Z \sim \text{Lap}(1/\epsilon n)$
- $|\tilde{\mu}_L - \mu| \leq \mathcal{O}\left(\frac{1}{\epsilon n}\right)$

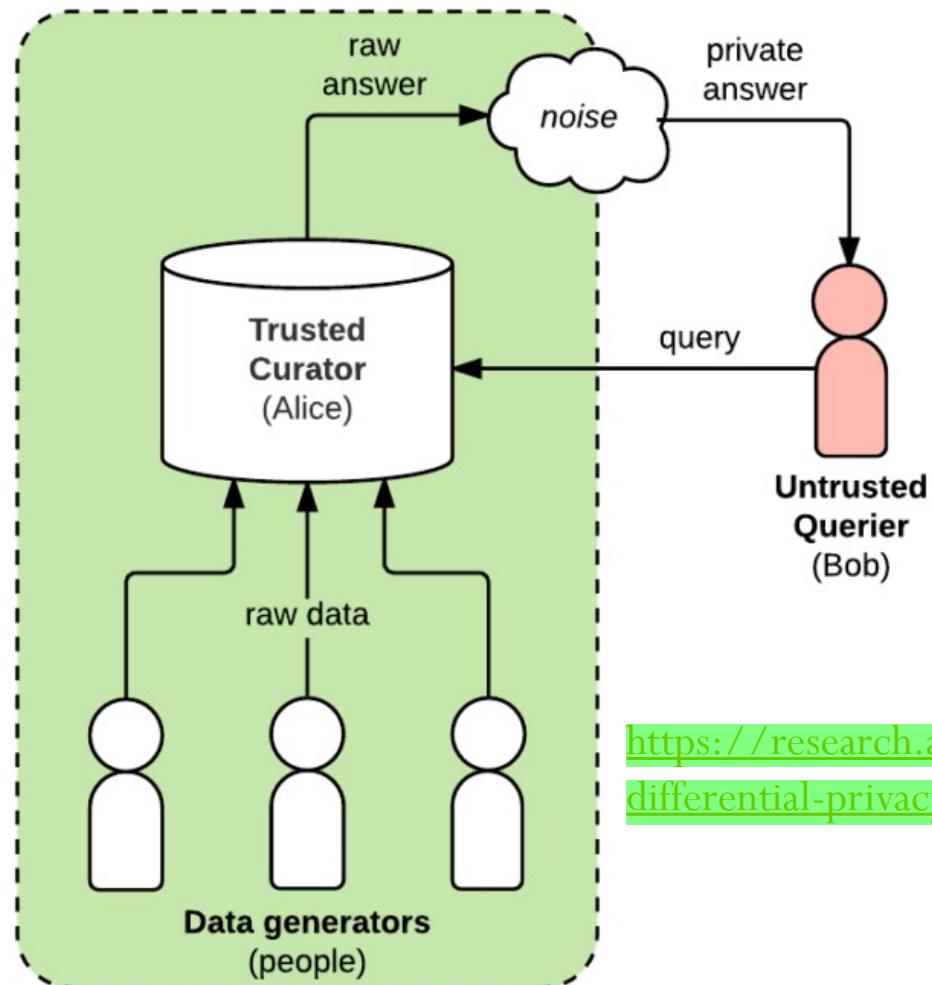
Randomized response vs Laplace

- Randomized response achieves local DP:
 - Each individual releases a **noisy** answer
 - **No trusted** curator with access to the raw data
 - $|\tilde{\mu}_R - \mu| \leq \mathcal{O}\left(\frac{1}{\epsilon\sqrt{n}}\right)$
- Laplace mechanism achieves global/centralized DP:
 - A **trusted** curator holds one bit $\mathbf{x}_i = \{0, 1\}$ for each of $\mathbf{n}$ individuals
 - Need a trusted central aggregator with access to the raw data
 - $|\tilde{\mu}_L - \mu| \leq \mathcal{O}\left(\frac{1}{\epsilon n}\right)$

Local vs global DP



Local privacy



Global privacy

<https://research.aimultiple.com/differential-privacy/>

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(ϵ, δ) -differential privacy

- $\mathbf{X}$: The data universe
- $\mathbf{D} \subset \mathbf{X}$: The data set (one element per person)
- Privacy parameters: $\epsilon \geq 0, 0 \leq \delta \leq 1$

Definition: An algorithm $\mathbf{M}$ is ϵ -differentially private if for *all* pairs of neighboring datasets $\mathbf{D}, \mathbf{D}'$, and for *all* outputs $\mathbf{R}$:

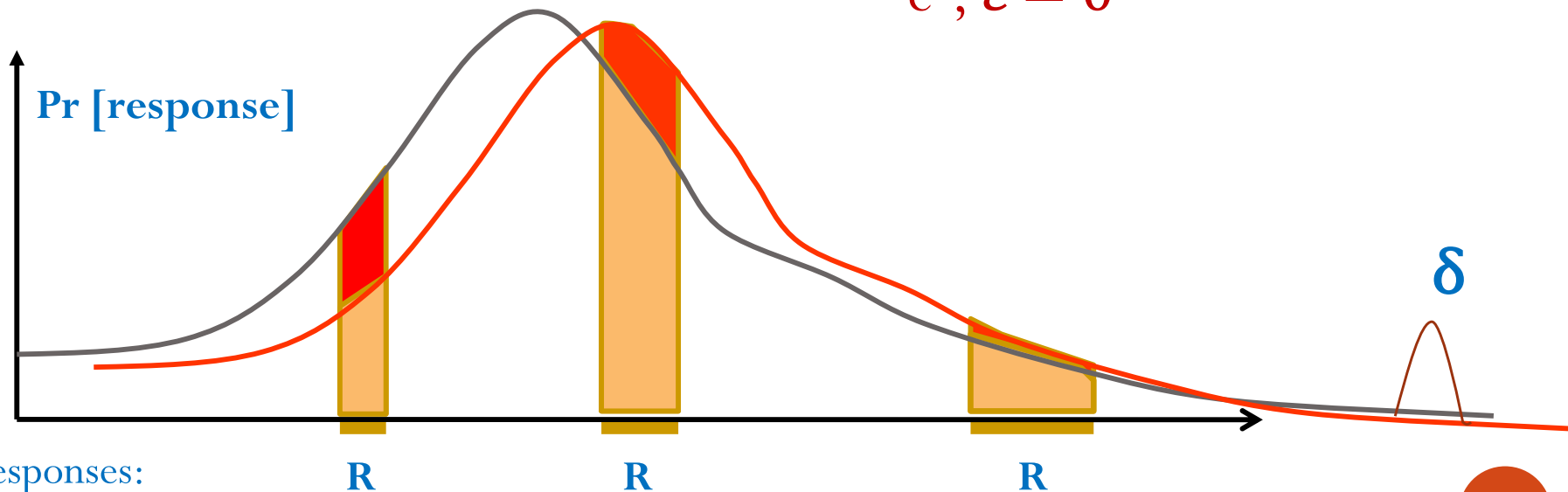
$$\Pr [\mathbf{M}(\mathbf{D}) = \mathbf{R}] \leq e^\epsilon \Pr [\mathbf{M}(\mathbf{D}') = \mathbf{R}] + \delta$$

- δ accounts for “bad events” that might result in high privacy loss

(ϵ, δ) -differential privacy

- Random sampling $M(x_1, \dots, x_n) = x_{\text{Unif}([n])}$ is $(0, 1/n)$ -DP
 - Consider $D' = D \ \& \ me$
 - $\Pr [\text{sample } x_{me} \text{ from } D] = 0$ and $\Pr [\text{sample } x_{me} \text{ from } D'] = 1/n$
 - $\Pr [\text{sample } x_{me} \text{ from } D'] \leq \Pr [\text{sample } x_{me} \text{ from } D] + 1/n$
 - $\Rightarrow$ We should take $\delta \ll 1/n$

$e^0, \epsilon = 0$



Fundamental properties

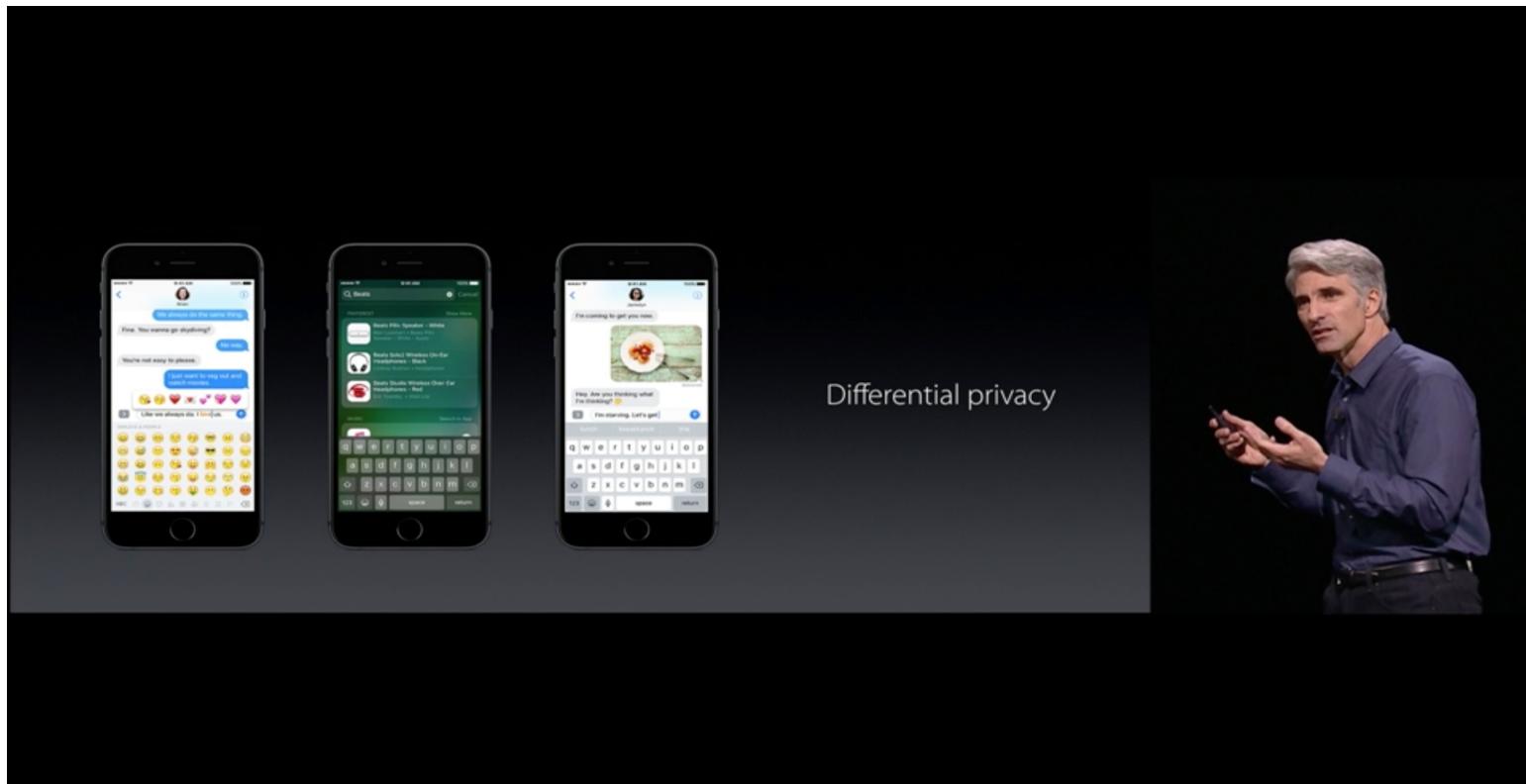
- Robustness to post-processing:
 - If $\mathbf{M}$ is (ϵ, δ) -DP then $\mathbf{F} \circ \mathbf{M}$ is (ϵ, δ) -DP
- Composition:
 - If $\mathbf{M}_j$ is (ϵ_j, δ_j) -DP for $j = 1, \dots, k$ then for a vector $\mathbf{x}$, the combination mechanism $\mathcal{M} = (\mathbf{M}_1(\mathbf{x}), \dots, \mathbf{M}_k(\mathbf{x}))$ is $(\sum_j \epsilon_j, \sum_j \delta_j)$ -DP. In the homogeneous case, this yields $(k \epsilon, k \delta)$ -DP.
 - This holds against active attacker (i.e. each query depends on the previous queries' result)
- Group privacy:
 - If $\mathbf{M}$ is (ϵ, δ) -DP w.r.t $\mathbf{D} \simeq^1 \mathbf{D}'$, then $\mathbf{M}$ is $(t \epsilon, t e^{t\epsilon} \delta)$ -DP w.r.t $\mathbf{D} \simeq^t \mathbf{D}'$ (i.e. t changes)

Outline

- Motivation
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 - RAPPOR
 - Private empirical risk minimization

Applications in big tech companies

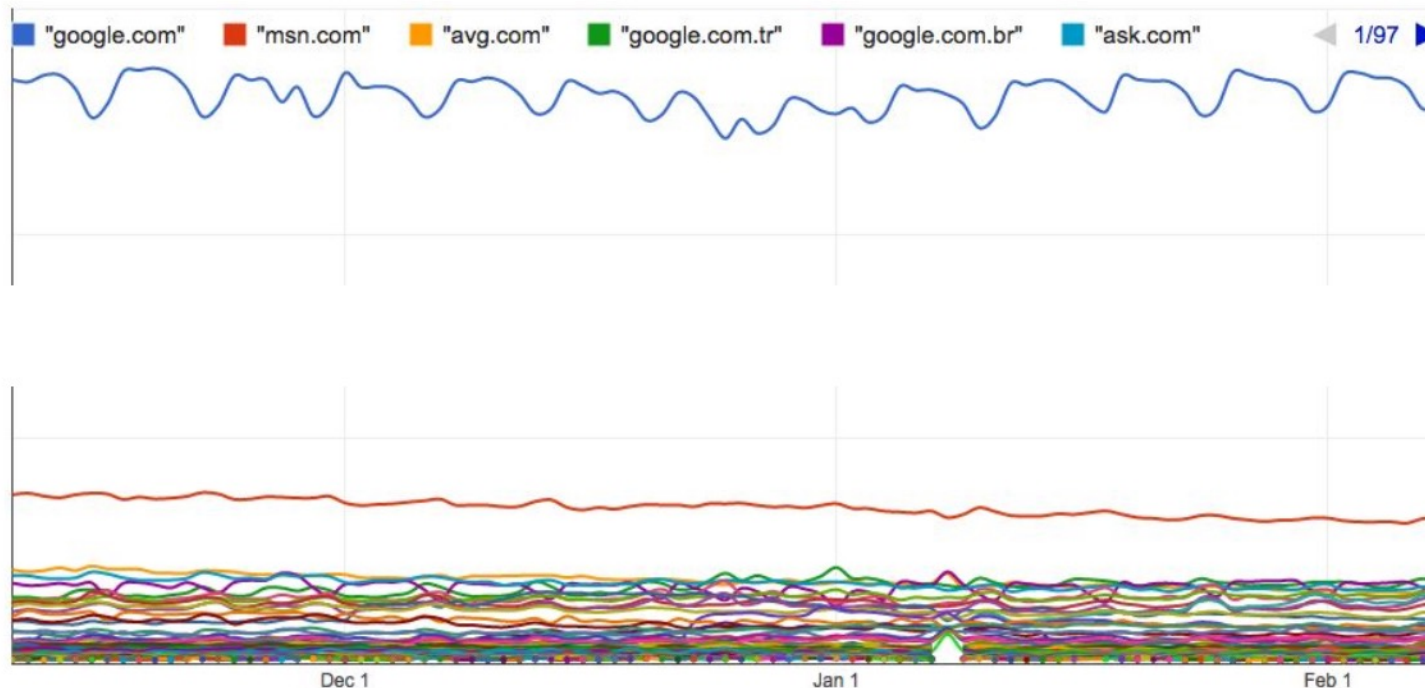
- In **Apple iOS** and **MacOS**, collect **typing statistics**
 - To replace text by an icon w.r.t semantics
 - To recommend the relevant text



Applications in big tech companies

- In Google Chrome browser, to collect browsing statistics
 - To estimate the popularity of Yahoo! Search, Bing, ...
 - To know the most Chrome homepage URL

Estimated proportions



google
msn
avg
google tr
google br

Applications in big tech companies

- In [Microsoft Windows](#) to collect telemetry data over time
- From [Snap](#) to perform modeling of user preference
- All deployments are based on RR, but extend it substantially
 - To deal with over 100 million users each
 - To handle the large space of possible values a user might have
- Local Differential Privacy is state of the art in 2018
 - Randomized response invented in 1965: five decades ago!

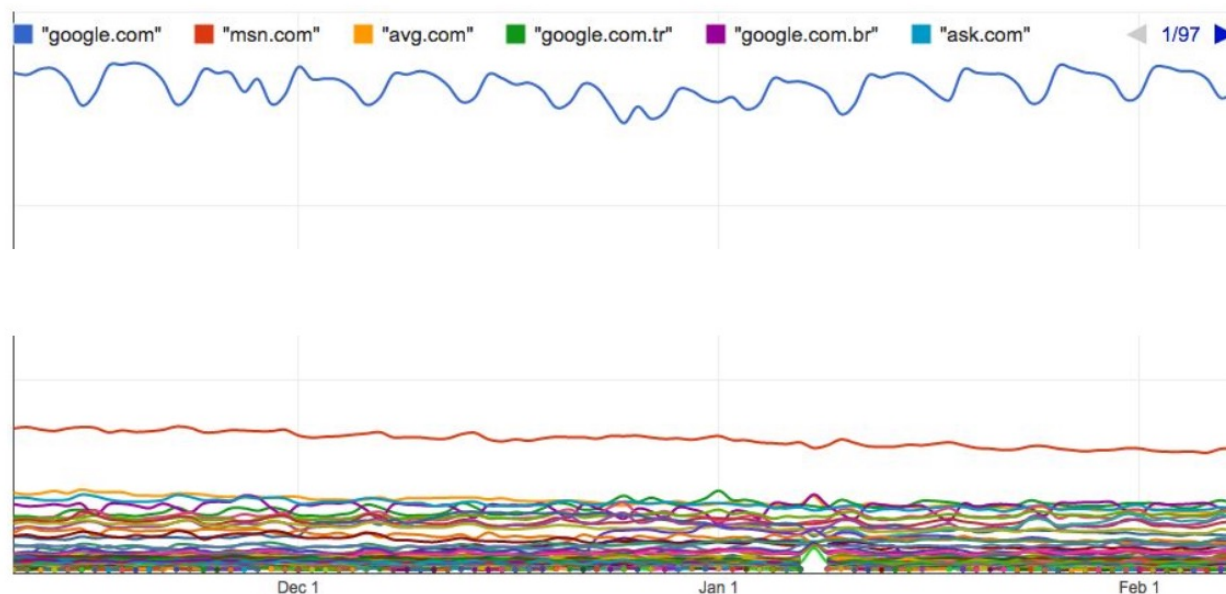
Recap of Randomized Response

- Each user has a **single bit** of private information
 - Encoding e.g. political/sexual/religious preference, illness, etc.
- **Randomize Response**: toss an unbiased coin [Warner 65]
 - If **Heads** (probability $p = \frac{1}{2}$), report the **true answer**
 - Else, toss unbiased coin again: if **Heads**, report **true answer**, else lie
- Collect responses from n users, subtract noise
 - Error in the estimate is proportional to $1/\sqrt{n}$
 - Simple LDP algorithm with parameter $\epsilon = \ln((\frac{3}{4})/(\frac{1}{4})) = \ln(3)$
 - Generalization: allow biased coins ($p \neq \frac{1}{2}$)

RAPPOR

- Deployed in Google Chrome to collect browsing statistics
- Use case: We want to know the most popular homepage
 - Each user has a 1 chrome homepage, e.g. google.com, yahoo.com, ...

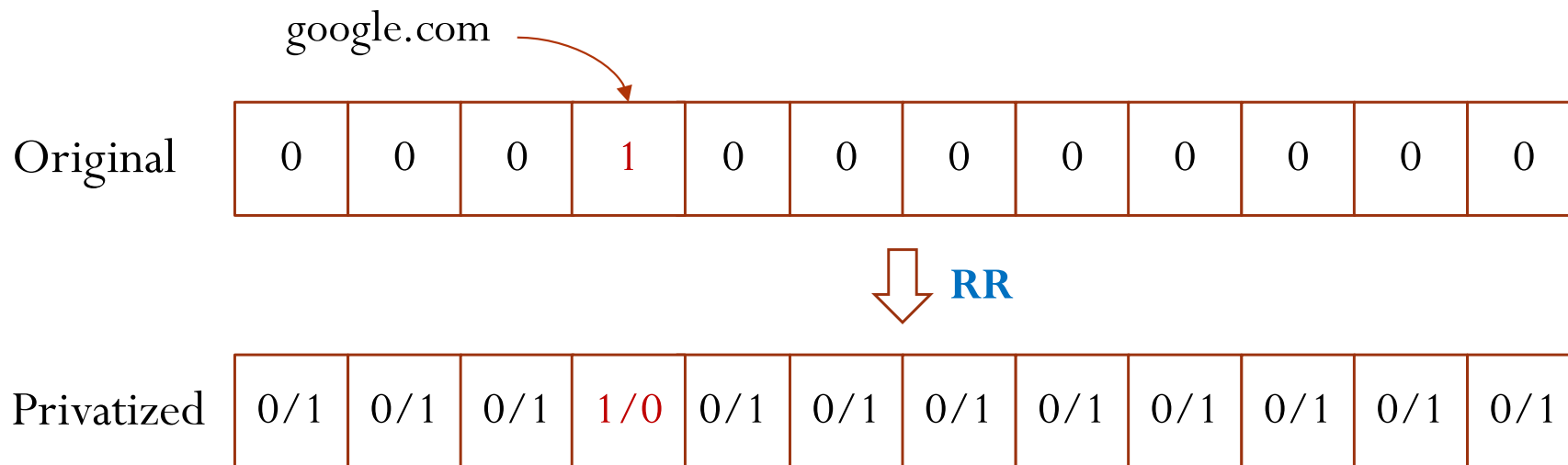
Estimated proportions



google
msn
avg
google tr
google br

RAPPOR

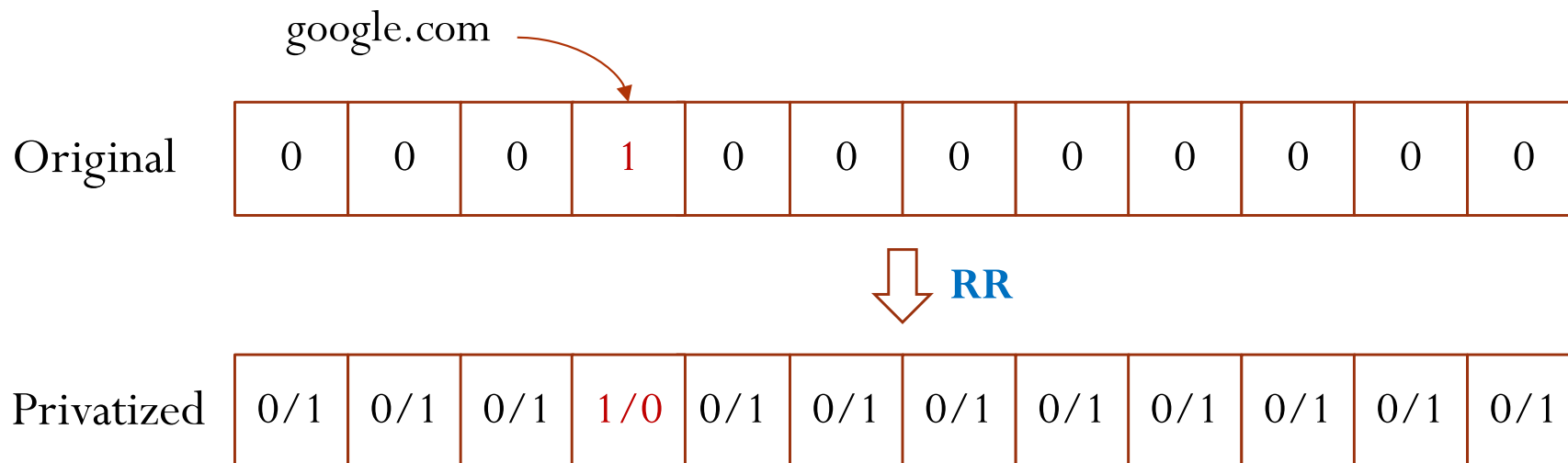
- First attempt:
 - Present all possible homepages (domain size) as a binary vector
 - A user has a sparse binary vector with one 1
 - Apply RR on every bit of the user vector



RAPPOR

- Limitations:

- **Slow:** Have to send each bit 1 of the privatized vector
- **Threat:** Average of repeated privatized vectors eventually reveals the answer

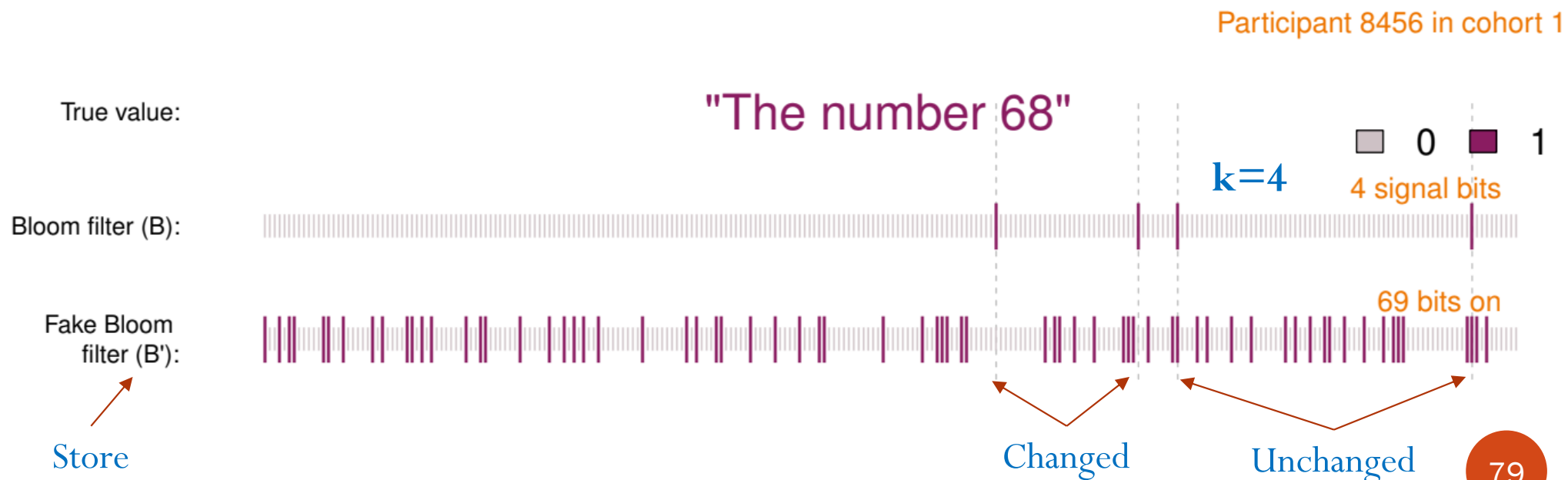


RAPPOR

- Limitations:
 - **Slow:** Have to send each bit 1 of the privatized vector
 - **Threat:** Average of repeated privatized vectors eventually reveals the answer
- Fix:
 - **Fast:** Reduce the domain size through hashing
 - **Privacy:** Memoize the RR-based privatized vector to reuse later, and apply a “random” flipping (e.g. $0 \rightarrow 1$ with prob. $1/2$, $1 \rightarrow 0$ with prob. $1/4$) on the privatized vector and report it

RAPPOR: 2-level randomization

- RAPPOR: Use a Bloom filter to reduce the domain size
 - Consider the URL as an integer (e.g. index of 1 in the binary vector)
 - Use k hash function to construct a Bloom filter $\mathbf{B}$
 - Apply RR on each bit of the Bloom filter to have a privatized $\mathbf{B}'$
 - Store the privatized Bloom filter $\mathbf{B}'$ to reuse later

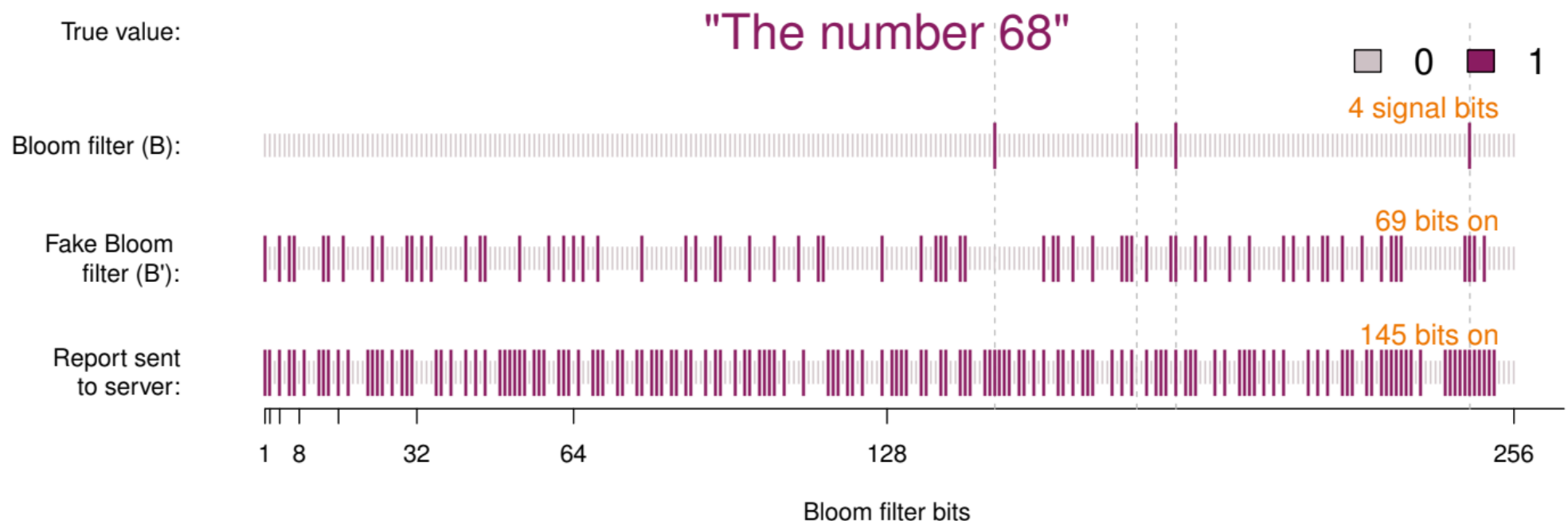


RAPPOR: 2-level randomization

- RAPPOR: Apply 2-level randomization on a Bloom filter

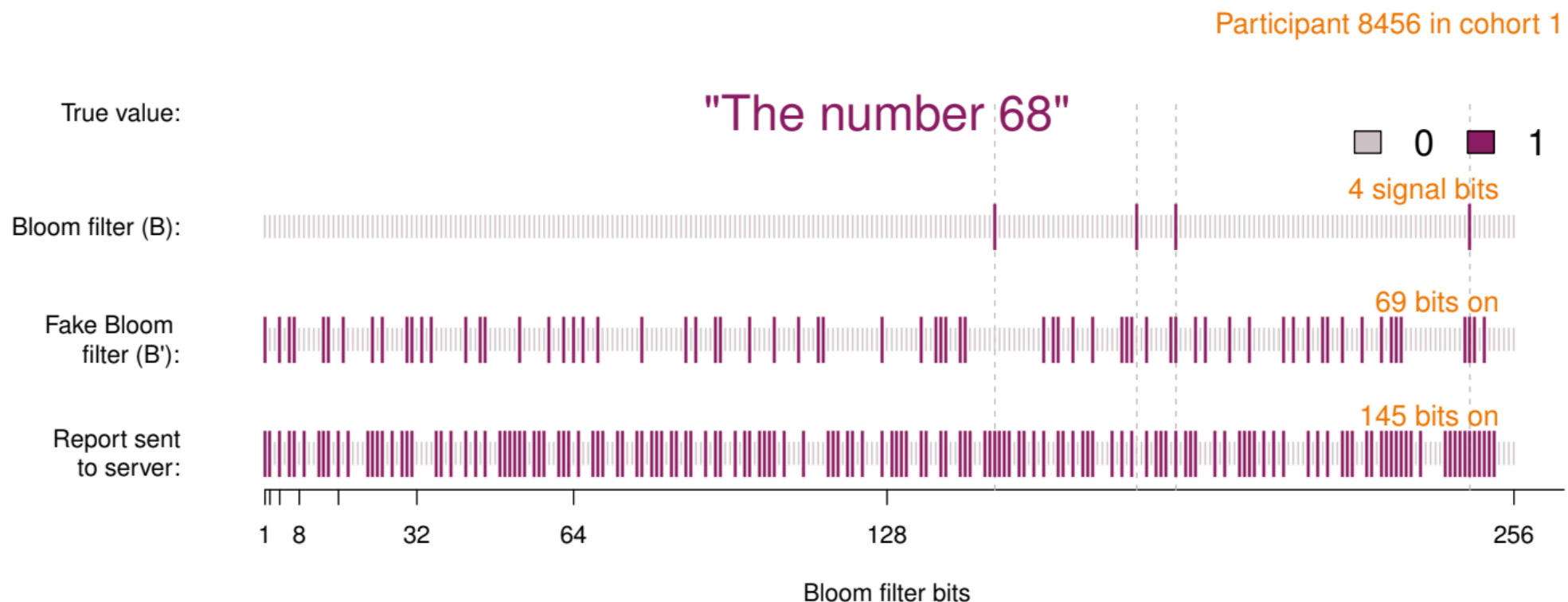
- Consider the URL as an integer (e.g. index of 1 in the binary vector)
- Use k hash function to construct a Bloom filter $\mathbf{B}$
- Apply RR on each bit of the Bloom filter to have a privatized $\mathbf{B}'$
- Apply random flipping on each bit of $\mathbf{B}'$ and send it

Participant 8456 in cohort 1



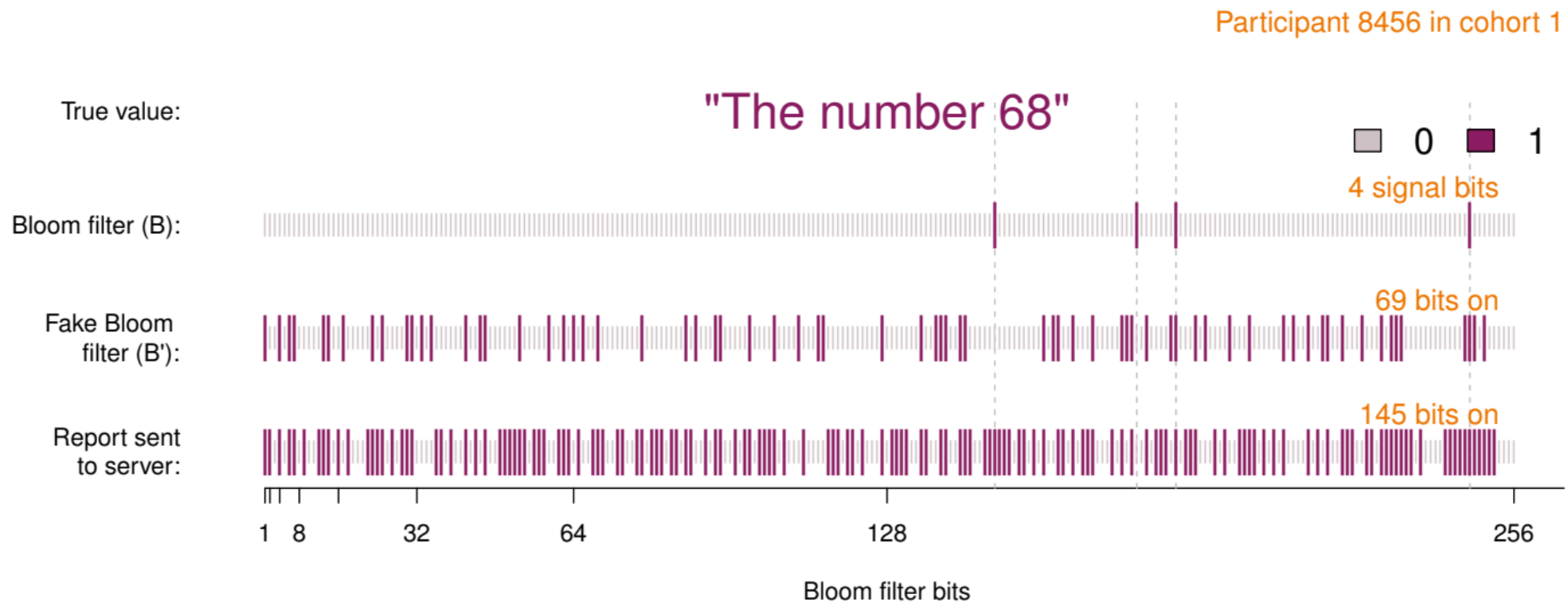
RAPPOR: 2-level randomization

- RAPPOR: Apply 2-level randomization on a Bloom filter
 - Storing a privatized $\mathbf{B}'$ to prevent that average of $\mathbf{B}'$ reveals true value
 - Apply random flipping on each bit of $\mathbf{B}'$ and send it to the server to prevent that repeatedly sending $\mathbf{B}'$ forms a unique tracking ID



RAPPOR: 2-level randomization

- RAPPOR: Apply 2-level randomization on a Bloom filter
 - If the RR is ϵ -DP, then $\mathbf{B}'$ is $k\epsilon$ -DP (i.e. protecting at most k bits)
 - The “random” flipping (e.g. $0 \rightarrow 1$ with prob. $1/2$, $1 \rightarrow 0$ with prob. $1/4$) ensure that the bit 1 of Bloom filter is well preserved



Decoding noisy Bloom filter & RR

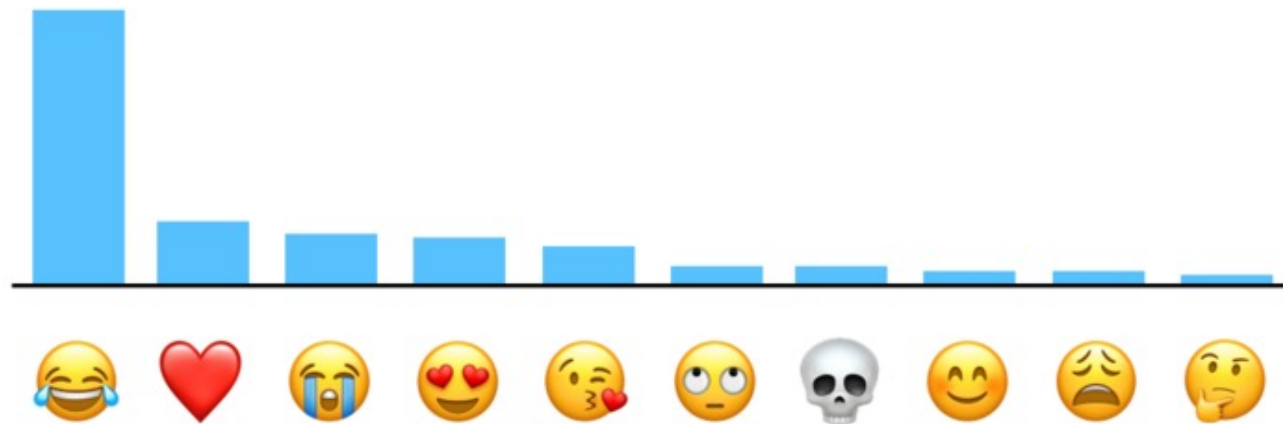
- Combine all user reports and observe how often each bit is set
 - Each report is a noisy Bloom filter
 - Each bit of the noisy Bloom filter is set with some probability
- To **estimate the frequency** of a particular value (e.g. URL):
 - Look up its bit locations in the Bloom filter
 - Compute the unbiased estimate of the probability each is 1
 - Take the **minimum** of these estimates as the frequency
 - More advanced decoding heuristics to decode **all** at once

RAPPOR in practice

- RAPPOR is implemented in the Chrome browser
 - Collect data from opt-in users, tends of millions per day
 - Open source implementation: <https://github.com/google/rappor>
- Tracks settings in the browser, e.g. homepage, search engine
 - Many users unexpectedly change homepage $\Rightarrow$ possible malware
- Typical configuration:
 - 128 bit Bloom filter, 2 hash functions, privacy parameter $\epsilon = 0.5$
 - Needs about **10K** reports to identify a value with confidence

Apple: Sketches and transform

- Similar problem to RAPPOR: count frequencies of many items
- Instead of Bloom filter, they use sketches (CountMean)
 - Similar idea but better suited to capturing frequencies



The Count Mean Sketch technique allows Apple to determine the most popular emoji to help design better ways to find and use our favorite emoji. The top emoji for US English speakers contained some surprising favorites.

Outline

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 - RAPPOR
 - Private empirical risk minimization

Private empirical risk minimization

- Setup:

- A curator has training data $\mathbf{X} = ((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n))$ about n individuals and wants to train a model by minimizing over $\theta \in \Theta$

$$L(X, \theta) = \frac{1}{n} \sum_{i=1}^n l(x_i, y_i, \theta) + \frac{R(\theta)}{n}$$

Objective function Loss function Regularization

- Examples: Logistic regression, SVM, linear regression, DNN, ...
- Problem: Curator wants to protect the privacy of $\mathbf{X}$

Private empirical risk minimization

- Setup:

- A curator has training data $\mathbf{X} = ((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n))$ about n individuals and wants to train a model by minimizing over $\boldsymbol{\theta} \in \Theta$

$$L(\mathbf{X}, \theta) = \frac{1}{n} \sum_{i=1}^n l(x_i, y_i, \theta) + \frac{R(\theta)}{n}$$

- Private empirical risk minimization (ERM) algorithms:

- **Output perturbation:** add some noise $\mathbf{Z}$ to $\hat{\boldsymbol{\theta}} = \operatorname{argmin}_{\boldsymbol{\theta}} l(\mathbf{X}, \boldsymbol{\theta})$
- **Objective perturbation:** Reveal the optimum of $l(\mathbf{X}, \boldsymbol{\theta}) + \langle \boldsymbol{\theta}, \mathbf{Z} \rangle$ for some noise $\mathbf{Z}$ (adding noise to the objective function prior to minimizing)
- **Gradient perturbation:** optimize $l(\mathbf{X}, \boldsymbol{\theta})$ using SGD with noisy gradients

DP-ERM comparison

Perturb	Optimization	Privacy	Assumptions	Excess Risk	Reference
Objective	Exact	(ϵ, δ)	linear model convexity	$\tilde{\mathcal{O}}\left(\frac{1}{\epsilon\sqrt{n}}\right)$	[Jain and Thakurta, 2014]
Output	Exact	(ϵ, δ)	linear model convexity	$\mathcal{O}\left(\frac{1}{\epsilon\sqrt{n}}\right)$	[Jain and Thakurta, 2014]
Output	SGD	ϵ	linear model convexity	$\mathcal{O}\left(\frac{d}{\epsilon\sqrt{n}}\right)$	[Wu et al., 2016]
Output	SGD	ϵ	linear model strong convexity	$\mathcal{O}\left(\frac{d}{\epsilon n}\right)$	[Wu et al., 2016]
Gradient	SGD	(ϵ, δ)	convexity	$\tilde{\mathcal{O}}\left(\frac{\sqrt{d}}{\epsilon n}\right)$	[Bassily et al., 2014]
Gradient	SGD	(ϵ, δ)	strong convexity	$\tilde{\mathcal{O}}\left(\frac{d}{\epsilon^2 n^2}\right)$	[Bassily et al., 2014]

See also [Talwar et al., 2014, Abadi et al., 2016]

Reference

- https://www.cs.utexas.edu/~shmat/courses/cs380s_fall09/
- <https://cs.ioc.ee/ewscs/2016/maffei/maffei-slides-lecture1.pdf>
- <http://dimacs.rutgers.edu/~graham/pubs/papers/ldptutorial.pdf>
- <https://sites.google.com/view/kdd2018-tutorial/home>
- [http://dimacs.rutgers.edu/archive/Workshops/BigDataHub/Slides/RAPPO
R-talk-for-DIMACS-workshop-April-2017.pdf](http://dimacs.rutgers.edu/archive/Workshops/BigDataHub/Slides/RAPPO
R-talk-for-DIMACS-workshop-April-2017.pdf)